

A Linear Clock for Random Walk on Tree-Weighted Planar Maps

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Abstract

We prove that stationary random walk on a planar map sampled with weight equal to its number of spanning trees converges to Liouville Brownian motion on the unit-area $\sqrt{2}$ -Liouville quantum sphere. The convergence retains the conditional path law jointly with the measured metric space. The walk chooses uniformly among all incident half-edges and starts from the stationary degree measure. For total attempt rate one and the continuum Dirichlet form with factor $1/2$, the time acceleration is exactly the number of map edges. The result uses the companion contour and metric limits for this same ensemble.

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1 Introduction

A scaling limit for the metric and area of a random surface does not by itself determine the motion of a particle on that surface. The electrical conductances of the approximating graphs can change the limiting diffusion, even when their measured metric spaces converge. A further question is whether the time needed to see the continuum motion has a single deterministic normalization.

We resolve this question for planar maps weighted by their numbers of spanning trees. The geometric input is the metric-measure limit and the contour reconstruction theory of [22]. We prove the additional convergence of the conditional walk laws, with an exact linear clock. The proof separates the identification of the electrical energy from the control of the speed measure. This separation is useful because neither part follows from metric-measure convergence alone.

1.1 The model and the continuum process

A rooted planar map is a finite connected graph embedded in the oriented sphere, up to orientation-preserving homeomorphism, with a distinguished oriented edge. Loops and multiple edges are allowed. For $n \geq 1$, let (M_n, T_n) be uniform among rooted maps with n edges decorated by one of their spanning trees. Thus the marginal weight of M_n is proportional to its number of spanning trees; conditional on M_n , the tree T_n is uniform. Write V_n for the vertex set and d_n for graph distance using all map edges, each of length one. Set

$$\mu_n = \sum_{v \in V_n} \frac{\deg(v)}{2n} \delta_v, \tag{1.1}$$

where each loop contributes two to the degree.

Conditional on the map, let X^n start from μ_n . At the times of an independent rate-one Poisson process it chooses an incident half-edge uniformly and moves to its other endpoint.

Choosing a loop makes no displacement. Its generator is

$$(L_n f)(v) = \frac{1}{\deg(v)} \sum_{e \text{ a half-edge at } v} (f(\text{other}(e)) - f(v)). \quad (1.2)$$

The measure μ_n is stationary and reversible for this walk.

Let (S, h) be an ordinary unit-area $\sqrt{2}$ -Liouville quantum sphere, with its distinguished points forgotten. We use the ordinary fixed-area sphere law of Duplantier, Miller, and Sheffield [5]. Its quantum area measure μ_h has total mass one. Write D_h for its Liouville quantum gravity metric in a fixed deterministic normalization. Identify the underlying conformal sphere with the round sphere. The continuum process B^h is the conservative μ_h -symmetric diffusion associated with the closure on $L^2(\mu_h)$ of

$$\mathcal{E}_h(f, g) = \frac{1}{2} \int_{S^2} \langle \nabla f, \nabla g \rangle_{\text{round}} d \text{vol}_{\text{round}}, \quad f, g \in C^\infty(S^2). \quad (1.3)$$

The closed form uses quasi-continuous representatives. This is Liouville Brownian motion, started with conditional law μ_h ; see [8, 2] and the Dirichlet-form framework in [7]. Conformal invariance of the two-dimensional Dirichlet integral makes (1.3) independent of the conformal coordinate. Its factor $1/2$ fixes the continuum clock.

For precision we fix the spatial normalization used throughout. For a finite real random variable Y , let $m(Y)$ be the unique real solution of $\mathbb{E}[\arctan(Y - m(Y))] = 0$. Put

$$\Delta_n = \max\{1, \text{diam}(V_n, d_n)\}, \quad \Delta = \text{diam}(S, D_h), \quad a_n = \exp\{m(\log \Delta) - m(\log \Delta_n)\}. \quad (1.4)$$

Both centers use unconditional laws, so a_n is deterministic. We use the geometric limit of [22] in the form

$$a_n \longrightarrow 0, \quad (V_n, a_n d_n, \mu_n) \xrightarrow{d} (S, D_h, \mu_h) \quad \text{in Gromov-Hausdorff-Prokhorov topology.} \quad (1.5)$$

The contour and topological results from that work needed in addition to (1.5) are stated in Section 2.

1.2 Retaining the conditional path law

Fix $T > 0$. If (K, d, μ) is a compact metric probability space, let Q be a Borel probability measure on $D([0, T], K)$. We regard (K, d, μ, Q) up to isometries transporting both measures. For two such quadruples, define δ_T as the infimum over isometric embeddings $i : K \rightarrow Z$ and $j : K' \rightarrow Z$ into a common complete separable metric space of

$$\max \left\{ d_H^Z(iK, jK'), d_P^Z(i_*\mu, j_*\mu'), d_P^{D([0, T], Z)}(i_*Q, j_*Q') \right\}. \quad (1.6)$$

Here d_H and d_P denote Hausdorff and Prokhorov distances. The last term uses the Skorokhod J_1 topology: time changes are increasing continuous bijections of $[0, T]$, and both their uniform time displacement and the uniform spatial discrepancy tend to zero. Each of the three distances may be truncated at one. An embedding acts on a path at every time.

Theorem 1.1 (Linear clock and conditional laws). *For every finite $T > 0$, define*

$$Q_n^T = \text{Law}((X_{nt}^n)_{0 \leq t \leq T} \mid M_n, T_n), \quad Q_h^T = \text{Law}((B_t^h)_{0 \leq t \leq T} \mid h).$$

With the spatial normalization (1.4),

$$(V_n, a_n d_n, \mu_n, Q_n^T) \xrightarrow{d} (S, D_h, \mu_h, Q_h^T) \quad \text{for } \delta_T,$$

through all positive integers n . Thus the deterministic clock multiplier for (1.2) and (1.3) is $c = 1$.

The random probability law is part of the state in Theorem 1.1. For example, the same argument gives joint convergence of any fixed number of conditionally independent stationary walks on the same map: in a common embedding, the associated conditional joint laws are the finite products of Q_n^T . Their weak convergence follows from that of Q_n^T . This preserves information about the shared environment.

1.3 Context and related work

Mullin’s enumeration of tree-rooted maps [20] gives the combinatorial starting point: the selected spanning tree and its complementary dual tree encode a map by a two-coordinate walk. The continuum counterpart is the mating-of-trees construction of Duplantier, Miller, and Sheffield [5]. At $\gamma = \sqrt{2}$ its boundary-length coordinates are independent Brownian motions. Miller and Sheffield [19] give the corresponding finite-area sphere description, and their imaginary-geometry construction [18] supplies the space-filling SLE_8 between the two continuum trees. Couplings of discrete and continuum encodings have also been used to compare graph distances across random-map models [10]. The continuum metric itself is characterized in [11], with conformal covariance established in [12].

Liouville Brownian motion was constructed by Garban, Rhodes, and Vargas [8] and independently by Berestycki [2] through a change of Brownian time determined by quantum area. Its Dirichlet-form description [9] separates two structures: the energy is conformally invariant, while the speed measure is quantum area. This distinction explains why the clock is a separate part of a random-map scaling limit.

Several works establish important parts of the discrete picture. Gwynne, Miller, and Sheffield prove convergence of the Tutte embedding of mated-CRT maps, including conditional walk-law convergence modulo time parameterization [15]. Their estimates for harmonic functions compare discrete and continuum energies up to constants and control oscillation [14]; their invariance principle for ergodic scale-free environments develops a broader framework for embedded networks [16]. Gwynne and Miller obtain resistance, return-probability, and displacement estimates for several infinite random planar maps, including spanning-tree-weighted maps [13].

Berestycki and Gwynne prove parameterized, quenched convergence to Liouville Brownian motion for mated-CRT maps [3]. Their deterministic time scale is an annealed exit-time median, comparable to the inverse quantum area of a cell. Whether its ratio to that inverse area converges is left open in [3, Remark 1.3]. Theorem 1.1 concerns the finite combinatorial tree-weighted map and its stationary half-edge walk. It retains the conditional path law together with metric and area, and identifies the acceleration as exactly n for (1.3).

Our geometric inputs are the companion results in [22]. The nondegeneration argument adapts the local-flow method in [21, Section “The canonical conformal structure”] to a primal

and a dual electrical network. The use of planar duality to determine an effective conductivity has classical precedents, including Dykhne's exact-conductivity argument [6]. Here we must first construct the energy limit without regularity assumptions on the shapes of discrete faces. A separate Green estimate in the original finite-map law then controls sources of small stationary mass and supplies the missing clock information.

1.4 Proof strategy and reusable estimates

For a finite graph embedded in the sphere, write

$$\mathbf{e}(u) = \sum_{ab \in E} (u(a) - u(b))^2, \quad (1.7)$$

with one term for each unoriented edge, including multiplicities. Loops contribute zero. The same convention defines $\mathbf{e}^*$ on the dual graph. The two main tasks are to identify the limit of these unnormalized energies and to pass from that limit to motion in actual time.

The contour description of the map gives local finite graph pieces in conformal coordinates. Section 2 explains why the local reconstruction and conditional-kernel calculations of [22] also apply to extremal lengths and constrained energy minima on these pieces. Exact edge incidences matter here: approximation of graph distances alone would not identify an energy. The extension retains compact numerical outputs before extracting a subsequence, and uses a change of measure depending only on the contour inputs.

Section 3 proves bounded-energy cutoff estimates without geometric control of individual faces. Its central argument is a planar obstruction to simultaneous degeneracy of the primal and dual networks. If annular extremal lengths could vanish, low-energy flows would have a nonzero limiting average length measure. Locality and triviality of field point germs produce transverse primal and dual crossing flows at a typical point. Every such pair crosses an edge and its dual, contradicting the vanishing traffic norms. The cutoffs then yield harmonic compactness.

Section 4 identifies every local variational limit as $\lambda \int |\nabla u|^2$, with the same positive deterministic λ for the primal and dual. To determine λ , we compare a condenser on an annulus with the period of its rotated current. The comparison in one direction gives $\lambda \leq 1$; recovery of an angular cochain and the signed primal–dual pairing give $\lambda \geq 1$. Thus the coefficient is exactly one in every subsequence. Section 5 transfers the result to fixed-area spheres using interior excursion densities and removes the single omitted contour endpoint by zero Sobolev capacity.

The remaining speed-measure issue is addressed by the following estimate, proved in Section 6. Let H_n be the positive unweighted graph Laplacian. For $\mu_n(f) = 0$, define $G_n f$ by $H_n G_n f = \mu_n f$ and $\mu_n(G_n f) = 0$, where the product $\mu_n f$ on the right denotes its vector of vertex masses. For every $0 < \rho < 1/2$ there are map-measurable random variables A_n with $\sup_n \mathbb{E} A_n < \infty$ such that, simultaneously for $\|f\|_\infty \leq 1$,

$$\|G_n f\|_\infty \leq A_n \mu_n(|f|)^\rho. \quad (1.8)$$

Its proof is entirely in the fixed-size map law. A spanning-forest identity expresses the centered inverse Laplacian through tree cuts; the contour bijection expresses each cut size as one plus a dual-tree distance. Dyck-excursion estimates then control the interval sums. This estimate can be used independently of the energy identification.

Finally, Section 7 uses (1.8) to obtain uniform resolvent convergence and a stopping-time tightness estimate. The resolvents identify the conditional path laws by their integrated finite-

dimensional distributions. The normalization is visible already at the discrete level:

$$-\langle f, nL_n f \rangle_{L^2(\mu_n)} = \frac{1}{2} \mathbf{e}(f).$$

Combining this identity with the coefficient $\lambda = 1$ gives the clock in Theorem 1.1.

2 Sewing and local electrical observations

Electrical energy depends on every edge, including parallel edges. Consequently, a description that approximates graph distances does not by itself suffice for our purpose. This section explains how the local sewing construction of [22] retains exact graph pieces and gives conditional laws for numerical observations on those pieces. The observations may be extremal lengths or constrained energy minima; we do not yet assert that they are bounded or identify their limits. The construction has three parts. We recover complete finite graph pieces from contour cells, pass their numerical observations to conditional limiting laws, and then read those laws from the fields in the corresponding spatial regions.

Throughout this section,

$$\gamma = \sqrt{2}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2} = \frac{3}{\sqrt{2}}, \quad \chi = \frac{1}{\sqrt{2}}.$$

All quantum surfaces use the field and boundary-length normalizations of [22]. A compact containment $K \Subset U$ means that K is compact and contained in the open coordinate domain U . A buffer for K is an open neighborhood whose closure is compactly contained in U .

2.1 The discrete graph and its continuum background

Let $Z^{\mathbf{d}} = (L^{\mathbf{d}}, R^{\mathbf{d}})$ be a bilateral walk whose increments are independent and uniform on the four axial unit vectors. For integer times $i \leq j$, set

$$i \sim_L j \iff L_i^{\mathbf{d}} = L_j^{\mathbf{d}} = \min_{i \leq k \leq j} L_k^{\mathbf{d}}. \quad (2.1)$$

The equivalence classes are the primal vertices. Horizontal steps give tree edges between consecutive classes, with the two traversals of a tree edge counted as one edge. A vertical up-step ending at u and the vertical down-step ending at $v > u$ are matched when

$$R_{u-1}^{\mathbf{d}} = R_v^{\mathbf{d}} < \min_{u \leq k < v} R_k^{\mathbf{d}}. \quad (2.2)$$

Each matched pair contributes one further edge between $[u-1]_L$ and $[v-1]_L$. Distinct matched pairs remain distinct edges when their endpoints agree. In particular loops and parallel edges are retained. Interchanging $L^{\mathbf{d}}$ and $R^{\mathbf{d}}$ gives the dual graph and its complementary spanning tree.

For a finite interval of indices, the *internal graph* uses the same vertex relation, all horizontal steps in that interval, and the vertical pairs whose two occurrences lie in the interval. Its vertices embed injectively into the full graph: the minimum test between two retained times is the same in both constructions. It need not contain every full-graph edge between those vertices.

Proposition 2.1 (Sewing and continuum background). *The following couplings and normalizations will be used.*

1. A uniform rooted map–spanning-tree pair with n edges is encoded by the axial walk conditioned to remain in $\mathbb{Z}_{\geq 0}^2$ and return to zero after $2n$ steps. The preceding rules recover every primal and dual edge with its incidences.
2. Give the bilateral walk an independent Poisson clock of rate m and multiply its heights by $m^{-1/2}$. As $m \rightarrow \infty$, its interpolated contour converges locally uniformly in distribution to $Z = (L, R)$, whose independent Brownian coordinates have variance $1/2$ per unit time. The limiting contour has the peanosphere coupling with a $\sqrt{2}$ -quantum cone h and an independent whole-plane space-filling SLE_8 , subsequently parameterized by μ_h . Write this curve as η and normalize $\eta(0) = 0$.
3. The imaginary field $\hat{h}$ is independent of h and is a whole-plane GFF modulo $2\pi\chi\mathbb{Z}$, with uniform phase in one period. It determines the oriented curve before area parameterization. The fields, curve, and contour are retained with their joint peanosphere law.
4. A finite interval of contour increments determines its abstract decorated quantum surface, its restricted curve, and the quantum lengths of its incoming and outgoing frontiers, modulo conformal coordinates. The analogous statement holds for every symmetric interior interval of the Brownian-excursion encoding of the ordinary unit-area sphere.

The discrete assertion is Mullin’s encoding [20], in the form of Theorem “Mullin encoding and its infinite version” in [22]. The continuum assertions are recorded in its Theorem “Peanosphere and interval surfaces” and its imaginary-field normalization. Their principal continuum inputs are the mating of trees theorems of [5, Theorems 1.9 and 1.11] and the space-filling construction of [18]. The use of a Poisson clock makes the two coordinate jump processes independent; index time divided by m has the same limiting area time.

We recall the geometric information accompanying this coupling. Two continuum times have the same image under η precisely when they are equivalent under the relation generated by

$$L_s = L_t = \inf_{[s,t]} L \quad \text{or} \quad R_s = R_t = \inf_{[s,t]} R, \quad s \leq t. \quad (2.3)$$

Except for the common endpoint in the sphere encoding, every class has at most three elements and belongs to one coordinate when it is nonsingleton. In a class $t_1 < t_2 < t_3$, the middle time is a strict local minimum; distinct strict local minima have different heights. At parameter 8, the interval cells are closed topological disks. The cone curve is proper, so inverse images of compact sets are bounded. These facts, including their simultaneous versions on a countable collection of intervals and independent Poisson cuts, are the topological input accompanying Theorem “Peanosphere and interval surfaces” of [22]; see also the identification description in [5, Section 1.3.1 and Lemma 8.15].

All subsequential arguments below can be placed in a coupling with locally uniform contour convergence. We attach the full continuum background conditionally on the limiting contour, independently of the additional *raw* discrete observations given that contour. Here *raw* means that the observation uses contour labels, without projecting them through η . This convention is compatible with further extraction. Indeed, for a bounded background test F , its conditional expectation $k_F(Z)$ can be approximated in L^1 of the fixed Brownian marginal by bounded continuous functions of Z . Multiplying by any bounded *raw* test leaves the approximation error uniformly bounded by the same L^1 error. Joint weak convergence therefore preserves the conditional sampling identity. Spatially projected observations will depend on both the raw data and the background; we only make this conditional sampling assertion before projection.

2.2 Projections and planar topology

Regard the map together with its dual as a cell decomposition of a surface, subdivided into triangles at its corners. Assign a primal or dual vertex the point $\eta(t)$ corresponding to any of its visits. All choices at one vertex, and all corners of an incident triangle, have uniformly vanishing discrepancy. Otherwise a subsequence of visit times would limit to a discrete equality or matching relation whose endpoints have distinct continuum images, contradicting (2.3). Join nearby assigned points by short coordinate arcs and fill the triangles in small disks. This gives continuous projections π_m with vanishing projected triangle mesh. On the plane this construction is made in a bounded window containing the region under consideration, after preserving its necessary stars and capping the window. This protection is obtained directly from lower contour visits, as in the proof below; it does not use a local instruction graph or any numerical-array limit.

Proposition 2.2 (Topological comparison). *In the contour coupling, the projections π_m may be chosen so that the following conclusions hold on the sphere and on every buffered compact part of the plane. Each conclusion is eventual for each fixed choice of margins.*

1. *If the projections of two primal vertices approach each other, they can be joined by primal paths with arbitrarily small projected diameter. This is uniform over all lifts in a fixed compact region. The same statement holds on the dual graph and for surface paths. In particular fixed tubes with clearance contain paths following them on either graph.*
2. *All lifts of a sufficiently small closed disk or other compact well can be included in a connected set over any fixed open enlargement. A loop projecting into a small disk can be filled over a slightly larger disk, provided that both have the prescribed clearance.*
3. *The projection has degree $+1$ or -1 in a compact spherical completion. Robustly transverse primal and dual tube crossings intersect at an edge and its dual. Projected circuits with nonzero winding difference about two separated wells separate all the corresponding lifted fibers.*
4. *Integer winding differences about two wells agree upstairs and downstairs up to the common degree sign, when loops and wells have fixed separation. These statements retain repeated edge and face incidences.*

Proof. The construction and the first three assertions are the construction preceding Lemma “Uniform topological joining”, that lemma, Lemma “Exterior connection and small-loop filling”, Proposition “Degree and robust planar topology”, and Lemma “Separation of entire fibers” in [22]. We explain their applicability to both graphs and to the plane.

The hypotheses are uniform contour convergence, vanishing projected cell mesh, and approximation of the limiting fiber generators by discrete links with small projected connecting paths. For a Brownian excursion above a level, use last and first crossings of nearby levels on the two sides of an interior time. Their discrete analogues approximate the chord. At a triple, apply this on the two excursions on either side of its middle minimum. The fiber description above therefore supplies every generator required for topological joining. Interchanging the coordinates gives the same argument for the dual. Applying planar separation to the two embedded graph paths gives the mixed crossing assertion: a primal path and a dual path can meet only through a primal edge and its dual, with the usual corner subdivision making the intersection literal on the surface.

For a bounded plane region, choose a window containing all its visits and its fixed additional

star layers. Both contour coordinates make strictly lower visits on both sides of a sufficiently large window. Lemma “Protected capping of a bounded window” in [22] then supplies a spherical completion preserving all these incidences. Its continuum completion is absolutely continuous with respect to a Brownian excursion, which supplies the sphere quotient needed for the preceding topological results.

For the integer assertion, connect all lifts of each well away from the tested loop, using the first two assertions. A two-chain bounded by that loop has constant multiplicity along each such connected fiber. Projection multiplies the difference of its multiplicities at the two fibers by $\deg \pi_m = \pm 1$. This is exactly the equality of winding differences. Equivalently, one can use relative homology after a simplicial perturbation inside the fixed margins. $\square$

2.3 Finite instruction graphs and numerical tests

For a finite multigraph G , write

$$\mathbf{e}_G(u) = \sum_{ab \in E(G)} (u(a) - u(b))^2. \quad (2.4)$$

The sum has one term for each unoriented edge; loops contribute zero. An energy test is the infimum of (2.4) subject to finitely specified value constraints at vertices of a specified subgraph. We allow $+\infty$ for an infeasible test. An extremal-length test for a family $\mathcal{P}$ of graph paths is

$$\text{EL}_G(\mathcal{P}) = \sup_{w: E(G) \rightarrow [0, \infty)} \frac{(\inf_{P \in \mathcal{P}} \sum_{e \in P} w(e))^2}{\sum_{e \in E(G)} w(e)^2}, \quad (2.5)$$

with the usual conventions for empty families and constant paths. Only nontrivial families will be used in the crossing arguments.

The graph on which one computes these quantities must be specified before taking a limit. We use the finite instruction graphs of Lemma “Finite-band reconstruction” in [22], recalled next. Their inputs consist of finite strings and relative heights. This description will also show why the conditional-law proof applies to our numerical tests.

Definition 2.3 (Finite packages). A *master block* consists of two increment strings on a time interval $[0, \ell_b]$, starting at zero, together with a finite list of internal cuts. A finite package contains finitely many master blocks. For each coordinate $a \in \{L, R\}$ choose a forest on these blocks and record its height differences $o_{bd}^a = s_d^a - s_b^a$. The values s_b^a are thereby determined within each forest component up to one additive constant.

A test chooses finitely many *active windows*, each between recorded cuts of a master block. Each active window initially has its own copy of the contour vertices. Its starting heights and increment strings are obtained by restriction of its master block. Enable the ordinary sewing inside each active window. The additional cross-window links are specified by the band rules below. A package can retain several tests on the same master strings.

For an active window b , write z_b^a for its increment string, s_b^a for its starting height, and $m_b^a = \min z_b^a$. Its incoming and outgoing record arcs are the times satisfying, respectively,

$$z_b^a(t) = \min_{0 \leq u \leq t} z_b^a(u), \quad z_b^a(t) = \min_{t \leq u \leq \ell_b} z_b^a(u).$$

A band from window b to window d permits outgoing-to-incoming links with common absolute height H satisfying

$$\alpha \leq H - q \leq \beta, \quad H \leq s_w^a + \min_J z_w^a \quad (w, J \text{ in the cap list}). \quad (2.6)$$

Here $\alpha < \beta$ are fixed rational parameters, q is the height at a recorded anchor cut, and each J is a recorded subwindow. The forest supplies every relative height appearing in the rule. Each selected minimum is evaluated from the discrete string itself.

In the primal graph, an L -band identifies its permitted vertices. An R -band is a rule for pairs of oriented step occurrences: an outgoing up-step and an incoming down-step must carry the same unit height strip, and the *whole strip* must satisfy the band and cap conditions. Such a pair contributes an edge; it does not impose an R -vertex identification. The dual uses the exchanged convention. If a tree edge is represented in several retained pieces it is counted once after the corresponding vertex identifications: its unordered endpoint pair identifies it, since a tree has no parallel edges. Each additional matched-step edge is counted once for that matched pair, even when another such pair has the same endpoints.

The same active windows and band instructions continue to define an old test when cuts are added or master blocks are merged. In the merged string, old starting heights are evaluated at the old cuts; they are no longer independent offset parameters. A graph allowed to use the entire merged interval is a separate new test. In particular an old test cannot be recovered by merely confining paths in that larger graph, since the larger graph may use newly enabled edges witnessed in a formerly omitted gap.

For each instruction list we retain a countable family of numerical outputs. A vertex is selected when one of its visits lies in a finite union of intervals of relative time. Endpoints can be recorded cuts or rational fractions between cuts. Such selectors specify the subgraph, its ports, and any confinement condition. Rational value intervals at selected visits specify energy constraints; repeated visits impose simultaneous constraints on the same vertex. The output library includes both graphs, finite joint lists of tests, and tests recomputed on union graphs. Inapplicable tests receive a fixed default value. Every nonnegative output is compactified in $[0, \infty]$, for example by $x \mapsto x/(1+x)$. We can retain outputs divided by any prescribed positive deterministic factor b_m , including factors tending to zero.

This gives compact countable products of numerical observations. The master strings, durations, cuts, and offsets remain separate input coordinates. Along any sequence $m \rightarrow \infty$, one may therefore extract joint limits of all the raw tests and the locally convergent contours. The issue is now to relate these formal tests to complete pieces of the actual map.

2.4 Exact cores and the selector sandwich

Add independent Poisson cuts in area time, using nested refinements. A cell is the image of a consecutive cut interval under η . Only cells compactly contained in the current testing domain will be used. Their local data consist of their oriented traversals, quantum area parameterizations, and tagged incoming and outgoing frontier lengths, with repeated frontier occurrences retained. A *run* is a finite union of consecutive cells, viewed with its concatenated traversal and tagged incoming and outgoing frontiers. For any finite selection of cells, its maximal runs are the maximal consecutive groups in that selection.

Here is how these local data provide the relative heights used in a package. Fix a run b and one coordinate $a \in \{L, R\}$. For an initial segment of the run, the incoming frontier length is

the drop from its starting contour height to the minimum on that segment; the outgoing length is the rise from that minimum to its ending height. Their difference is therefore the contour increment. Reading these two tagged lengths at every rational relative area time, and using continuity, recovers the whole increment path z_b^a from the local traversal. Put $m_b^a = \min z_b^a$ and $e_b^a = z_b^a(\ell_b)$.

The incoming frontier of the whole run has relative-height range $[m_b^a, 0]$, and the outgoing frontier has range $[m_b^a, e_b^a]$. For a tagged point p on either frontier, define its *relative height* $\vartheta_b^a(p)$ by starting with value m_b^a at the minimum endpoint and increasing by quantum boundary length along that frontier toward its tip. Thus the incoming tip has height 0 and the outgoing tip has height e_b^a . This orientation is by increasing height, irrespective of the orientation in which the contour traverses the frontier. Equivalently, if p is represented by a record time t , then $\vartheta_b^a(p) = z_b^a(t)$.

A *directed shared seam* $b \rightarrow d$ is a common frontier subarc of positive quantum length, tagged as outgoing for b and incoming for d , in the same coordinate. At a paired point p of this seam, the two absolute heights agree. Hence

$$s_b^a + \vartheta_b^a(p) = s_d^a + \vartheta_d^a(p), \quad s_d^a - s_b^a = \vartheta_b^a(p) - \vartheta_d^a(p). \quad (2.7)$$

The right side is constant along the seam and is determined by the two local tagged frontier parameterizations. One may read it at an interior point, so the construction does not require selecting or assigning a height to a branch endpoint. Taking a spanning forest of such seams supplies exactly the offsets in Definition 2.3. Their outgoing and incoming tags also specify which linked run precedes the other. A chronological order between runs without a declared seam is unnecessary.

There is one endpoint configuration requiring an exact comparison. Suppose three visits of one coordinate meet at height H , in three disjoint runs b, w, d in that order, with the middle visit a strict local minimum. The outer runs b, d share a positive interval of seam heights below H . Above H , the two positive seam intervals are $b \rightarrow w$ and $w \rightarrow d$; see Figure 1. Formula (2.7) on these intervals recovers s_w^a relative to the outer starts. The tag directions identify w as the middle run. It lies between the two outer visits and contains the middle contact. The contour stays at least H throughout that interval by (2.3), and attains H in w . Hence

$$H = s_w^a + m_w^a. \quad (2.8)$$

A band for the outer pair may extend above the continuum value H , but it must include the cap in (2.6) evaluated as $s_{w,m}^a + \min z_{w,m}^a$ at every discrete resolution. It is this exact minimum, rather than a rounded approximation of H , that excludes the outer-to-outer links above the branch.

We use bands lying inside these shared height intervals with positive margins at their effective endpoints, with two exceptions prescribed by the record data. At a natural lower record minimum the rational band starts below the minimum and is clipped by the exact record condition. At an upper endpoint with the three-seam configuration, the band is clipped by the exact selected minimum (2.8). These are the admissible band choices in the proof of Lemma “Finite-band reconstruction” of [22]. All their tests use the local tagged frontiers, relative increments, and quantum lengths just described.

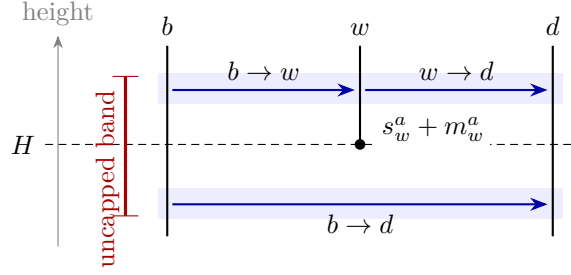


Figure 1: A triple contact in height coordinates. The shaded strips indicate positive intervals of shared quantum-boundary heights; they are not drawings of Euclidean seams. The outer pair is linked below H , while above H each outer run is linked to the middle run. The middle run’s exact discrete minimum caps the outer-pair band.

Lemma 2.4 (Complete graph pieces in a buffer). *Fix $K \subseteq U$ in a coordinate chart. Almost surely, there is a finite prescription, selectable from the local cells in U and their frontier data, with the following property. In a neighborhood of K , its instruction graph agrees with the actual discrete graph, including vertex equalities and all edge incidences, for all sufficiently large resolutions in the coupling. The same prescription can retain both the primal and the dual graph. One can require any fixed smaller collar of this neighborhood before restricting the tests to it.*

The prescriptions can be taken from the countable instruction scheme above and chosen measurably using the fields and spatial cut marks in U , with an arbitrarily small open buffer. The assertion also holds in the interior of a fixed interval cell, using only its internal increment string.

Proof. We use the exact-incidence part of the proof of Lemma “Finite-band reconstruction” in [22]. We give the argument because exactness, rather than an approximation of path costs, is essential for energy tests.

Choose $r > 0$ so that the closed $3r$ -neighborhood of K lies in U . Properness bounds all its visit times in a finite window. Refine the Poisson cuts until every cell meeting this neighborhood has diameter less than r . Select all cells meeting the closed $2r$ -neighborhood of K and merge consecutive selected cells into maximal runs. These runs are disjoint intervals; all visits sufficiently near K lie in them. Their endpoints stay away from K , since otherwise the neighboring cell would also have been selected. Local traversal data identify the selection: first inspect a slightly larger buffer and require the portions around every visit to the smaller compact to be flanked by marks while they remain in that buffer. Properness, continuity, and refinement make this condition hold eventually. No global index for the retained runs is part of this inspection.

Consider one coordinate and two selected runs $b < d$ in chronological order. An outgoing point of b and an incoming point of d at height H are linked exactly when

$$H \leq \min_{\text{between the two runs}} Z^a. \quad (2.9)$$

The same assertion holds discretely, with the full-strip convention for a matched step. Split the intervening interval into selected runs and omitted gaps. If a limiting link projects near K , any intervening time at its height represents that same continuum point. It must therefore belong to a selected cell. Thus every omitted-gap minimum is strictly above the height of that link. The relevant endpoint pairs form a compact set, and there are only finitely many gaps. Their positive

slack is consequently uniform: a sequence of slacks tending to zero would limit to a contact in an omitted gap, contrary to the selection.

An obstruction without slack can only be a minimum of a selected run. At a triple contact, the three directed seams in Figure 1 identify the middle run and recover its offset through (2.7). Formula (2.8), evaluated on the discrete string, supplies the cap in (2.6). At a natural lower seam endpoint, choose the lower rational band endpoint below the record minimum; the exact incoming or outgoing record condition then clips the band at that minimum. This retains links at the minimum as well as links at nearby lattice heights.

For every pair of runs, cover the compact set of relevant link heights by finitely many such bands. Choose each band to be admissible over its entire effective height interval: that interval lies in a genuine shared seam, its omitted-gap comparisons have positive slack, and every allowed non-strict cap is the exact selected minimum just described. The natural record conditions provide the exact clipping at lower seam endpoints. Include the positive seam intervals needed to read the relative heights, and retain a spanning forest of these comparisons in each coordinate. All needed offsets are sums along that forest. Seam directions specify the order of linked runs and distinguish each middle cap; the instructions do not need an order between unrelated runs. Each band retains every discrete link satisfying its rule. Uniform contour convergence, the positive omitted-gap slack, and the exact selected minima show two things: every actual link near K belongs to one of these bands eventually, and every permitted link in those effective bands is an actual link eventually. In particular the latter assertion applies to any permitted link with one endpoint over a smaller core collar: its two limiting endpoints lie on the same shared-seam point and hence remain inside the captured collar. The assertions hold simultaneously for all links, regardless of their number.

Vertex equality is the same first-coordinate minimum test, so it is captured as well. Start a chain of enabled identifications at a core representative. Its first link is an actual identification by the preceding band validity. All visits to that actual vertex have uniformly close projections, independently of the instruction graph, and lie in the captured collar. Inductively the next enabled link is therefore actual as well and remains at that vertex. Thus a peripheral chain cannot introduce an additional equality at the core. Horizontal tree traversals with the same identified endpoint pair give one tree edge; no two distinct tree edges have that pair. Added edges are instead identified by their matched step-occurrence pair, preserving distinct parallel edges. Hence the result is equality of graphs with incidences, not merely equality of their path relations. Applying the argument to the other coordinate gives the dual conclusion.

All seam, cap, and positive-margin conditions are measurable in the oriented local cells and frontier lengths. Rational bands, integer lower bounds on positive margins, and finite forest choices give a countable selection rule. Taking a further collar before the final restriction gives the stated neighborhood of K . Inside a fixed interval cell, every visit to a compact subset of its interior lies in that interval with a time margin; the same argument then uses only its internal string. $\square$

An arbitrary union of overlapping formal window copies need not be an injective copy of a map subgraph. The lemma asserts the existence of *complete* prescriptions, obtained from disjoint maximal runs or from a freshly computed merged graph. All comparisons to the actual map below use these complete prescriptions.

Lemma 2.5 (Spatial and value selectors). *In the setting of Lemma 2.4, suppose $K \subset O$ with K compact, O open, and $\overline{O} \in U$. There is an interval selector on a complete prescription which,*

eventually, includes every vertex projected into K and includes only vertices projected into O . Several such choices may be made together, for confinement and for each port set.

Let u be continuous on a neighborhood of $\bar{O}$. For any two uniform value tolerances $0 < \delta_- < \delta_+$, finite rational interval constraints on selected visit labels can be placed between the constraints $|v - u \circ \pi_m| \leq \delta_-$ on a slightly larger compact set and $|v - u \circ \pi_m| \leq \delta_+$ on a smaller one. All these comparisons hold simultaneously for every continuous target through the same retained test library.

Proof. On the finitely many selected windows, the limiting parameterizations are continuous. Cover the inverse image of K by finitely many relative-time intervals with closures projecting inside O . Their endpoints may be chosen from the prescribed dense countable relative-time library. Uniform projection accuracy and the complete collar of Lemma 2.4 give the desired inclusions. Select a vertex when any of its visits is in this union. The same argument applies to each port, while vanishing edge mesh allows an induced-edge rule within the complete prescription.

For value constraints, refine the intervals until the oscillation of $u \circ \eta$ on each is less than a fixed fraction of $\delta_+ - \delta_-$. Choose a rational value interval between the two allowed tolerances. Impose it at every selected visit in that interval, retaining all simultaneous constraints when a vertex has several visits. The strict tolerance gap and uniform projection error give the two implications. The construction chooses among an already countable collection of tests; it does not add a new exceptional null set for each target u . Compact exhaustion and rational positive tolerances therefore yield common measurable versions for all continuous targets. $\square$

The selector lemma compares variational values with strict margins. It does not assert convergence for a prescribed sharp boundary. For a crossing event, for example, choose smaller port and confinement constraints and then relaxed constraints, both strictly inside the desired geometry. An intermediate selector produces actual paths in the relaxed test, and any path family having the smaller clearance can be included by a sufficiently relaxed selector. Thus existence of a strictly fitting test with finite limiting extremal length is independent of the particular complete prescription. Exhausted energy values have the same property. These conventions also make readouts with a varying spatial center jointly measurable: all membership and value comparisons use finite selectors with a strict gap, followed by countable exhaustion.

2.5 The conditional law of enlarged outputs

We now identify the conditional law of the finite numerical arrays. The proof is the finite-experiment argument of [22], with the output vector enlarged before taking a subsequence. The enlargement is harmless for a specific reason: every change of density in that argument is a function of the inputs alone.

Lemma 2.6 (Conditional transfer for numerical outputs). *Consider finitely many packages as in Definition 2.3. Different packages use disjoint master blocks and disjoint offset forests. Sample their Poissonized strings independently; give their durations, cuts, and offset vectors smooth positive densities, with offsets discretized on the height lattice. Retain any finite or countable vector of compactified numerical tests of their instruction graphs, including tests on both graphs.*

Along a common subsequence the reference laws have disintegrations

$$\nu_r(dx_r)K_r(x_r, dy_r),$$

where x_r denotes the durations, cuts, increment strings, and forest offsets, and y_r denotes the numerical outputs. Insert these blocks in the actual bilateral walk with positive omitted gaps. Conditional on the entire limiting contour and its cuts, their output law is

$$\bigotimes_r K_r(x_r, dy_r). \quad (2.10)$$

The kernels are invariant under changes of positive reference densities and under relabeling blocks or changing the forest basis. They are compatible with jointly retained refinements and merged tests. These identities hold after selection from the countable instruction library using the limiting contour and cuts.

Proof. This is the argument of Lemma “Finite reference experiment and conditional transfer” in [22]. We include its measure calculation to specify its application to potentially discontinuous variational outputs.

Write $\epsilon_m = m^{-1/2}$. For one coordinate, order all selected blocks chronologically as $1, \dots, k$. Let s_i be their starts, e_i their net increments, and $g_i > 0$ the intervening durations. Complete the union of the package forests to a tree T on these blocks. Give its extra offsets independent positive densities. If the density on an edge e is ρ_e , use lattice mass

$$r_{m,e}(a) = \int_{a-\epsilon_m/2}^{a+\epsilon_m/2} \rho_e(u) du, \quad a \in \epsilon_m \mathbb{Z}.$$

Let $p_t(j)$ be the transition mass of one coordinate of the unscaled rate-one axial Poisson walk, so its variance rate is $\sigma^2 = 1/2$. The exact density of the actual relative starts with respect to these reference offsets is

$$W_m^a = \frac{\prod_{i=1}^{k-1} p_{mg_i}(\epsilon_m^{-1}(s_{i+1} - s_i - e_i))}{\prod_{e \in T} r_{m,e}(a_e)}. \quad (2.11)$$

Indeed, tree differences and consecutive differences are related by an integer linear map with integer inverse, obtained by summing along paths in T . Its determinant is ± 1 , so there is no lattice multiplicity. Multiply (2.11) for the two coordinates and include the density ratio for the durations and cuts to obtain W_m .

On compact input sets with all gaps bounded below by a positive constant, the local central limit theorem gives

$$\epsilon_m^{-1} p_{mt}(j) \longrightarrow (2\pi\sigma^2 t)^{-1/2} \exp\{-x^2/(2\sigma^2 t)\}, \quad x = \epsilon_m j, \quad \sigma^2 = \frac{1}{2},$$

uniformly in bounded x and in such t . The numerator and denominator of (2.11) have the same number of lattice factors. They cancel, leaving uniform convergence $W_m \rightarrow W$ to a continuous finite function on these compact sets.

Conditional on the selected inputs, the omitted strings are independent bridges and are independent of the numerical outputs. Their rescaled laws converge uniformly on these input sets to Brownian bridge laws. For completeness, their finite-dimensional densities are ratios of transition masses, so the preceding local limit theorem applies. On the first half of a bridge the density relative to the free walk is bounded by

$$\frac{\sup_j p_{m(t-u)}(j)}{p_{mt}(j_0)} \leq C, \quad u \leq t/2,$$

when t and $\epsilon_m j_0$ range in the chosen compact sets. The numerator is at most $C\epsilon_m$ and the denominator at least $c\epsilon_m$. Free-walk tightness and the reversed argument on the other half prove uniform bridge tightness. The same applies to a finite list of gaps and to exterior contour portions on a bounded window.

The full independent reference experiments converge jointly to the product of their limits, because their output spaces are compact and their input marginals converge. Let a denote the completion offsets and z the omitted bridges. Integrating bounded continuous bridge tests first, the preceding uniform convergence gives the limiting actual law

$$W(x, a) \nu(dx, da) \mathcal{B}(x, a; dz) \prod_r K_r(x_r, dy_r). \quad (2.12)$$

The weight and the bridge law depend only on contour inputs. The complete contour and cuts recover (x, a, z) , so disintegration of (2.12) gives (2.10).

To remove the compact restrictions, first keep all finitely many limiting gaps positive, then bound endpoints and offsets, and finally restrict the block paths to compact sets. Under the actual law the excluded probability tends to zero by finite-window contour tightness. If P_m is the actual law and Q_m the reference law, the exact identity

$$\mathbb{E}_{Q_m}[W_m \mathbf{1}_{A^c}] = P_m(A^c)$$

controls the omitted density mass. No global bound on W_m is required. Exhaustion, followed by a monotone-class argument, gives the assertion for bounded measurable tests and for the entire bilateral contour.

A change of reference density adds another input-only density ratio, continuous in the limit on compact sets, and thus leaves K_r unchanged. Forest bases are related by unimodular integer maps. The pushed-forward lattice cells and the directly discretized cells have the same volume and vanishing diameter. Their mass ratios converge to one on compact sets; tightness makes the two lattice laws converge in total variation. This comparison remains valid after applying a discontinuous graph-output map. Absolute time labels and the order of unrelated blocks occur only in W_m , and disappear from the conditional output law.

For merged tests one must also check the old output. Begin with the old independent blocks, their offsets, and their old numerical outputs. Insert the formerly omitted positive-duration gaps by the exact density and bridge construction. The resulting string has the new master-block law. Recompute the old graph from its unchanged active windows and bands, and compute the new union graph as a separate output on that same string. Formula (2.12), applied before conditioning on the new master string, proves the old conditional marginal. It never evaluates a previously chosen kernel on a forced-offset diagonal. Windows with a common endpoint were already within one maximal run, so no positive-density argument is applied to a zero-length gap. Jointly retaining finite vectors proves consistency for all refinements and unions, and then for the countable library.

Every step used only compactness of the output coordinates and their measurability from their own package. Thus it applies to the energy and extremal-length tests, with any deterministic normalization. It does not require continuity of those outputs or preservation of conditional independence under arbitrary weak limits. Finally, after conditioning the identities hold for all countably retained templates on one event; contour- and cut-measurable selection from them is therefore permitted. $\square$

2.6 Spatial locality and changes of quantum units

The preceding kernels are indexed by contour blocks. To use them in disjoint spatial regions, their inputs must be readable from the fields in those regions. This is where oriented cells and relative frontier heights enter.

Proposition 2.7 (Local numerical-array kernels). *On each open coordinate domain, take the limits of the numerical tests on complete buffered prescriptions, with relative labels localized to their cells. Conditional on $(h, \hat{h})$, the resulting local arrays admit kernels depending only on the fields in an arbitrarily small open enlargement of the testing region. Arrays using disjoint such enlargements are conditionally independent given the complete fields.*

One may retain the local Poisson marks. Their conditional laws are the corresponding Poisson processes with intensity proportional to μ_h , and their restrictions to disjoint domains are independent. The kernels are consistent under restriction, common refinements, and compact exhaustion.

Proof. Proposition “Background locality” in [22] states that $\hat{h}$ in a buffer determines the oriented curve segments through the testing region. The restriction of h gives their area parameterizations and their directed frontier lengths. These data determine the locally retained marked cells. For a segment from time s to time t , its coordinate increment is the outgoing frontier length minus the incoming frontier length of that coordinate, with tagged multiplicities retained:

$$Z_t^q - Z_s^q = \nu_h(\partial_q^{\text{out}} \eta([s, t])) - \nu_h(\partial_q^{\text{in}} \eta([s, t])), \quad q \in \{L, R\}. \quad (2.13)$$

Formula (2.13) reads the increments without an absolute contour height. Positive shared seams read the forest offsets, and the three-seam configuration in Lemma 2.4 reads the offsets needed at a minimum cap. Durations and relative cuts are local area data. Thus all four types of input in Lemma 2.6 are local.

For finitely many prescriptions in disjoint spatial domains, merge within each domain. Different domains use disjoint cells; two runs from different domains cannot share a cut, since its image would belong to both compactly contained cells. Their intervening chronological gaps are therefore positive even if the runs interleave in time. Lemma 2.6 gives their product law conditional on the contour and cuts. Its invariance under labels and forest bases lets us enumerate the cells locally without retaining a global cut index.

Adjoin the continuum background by the conditional sampling convention above. The product formula remains valid upon conditioning additionally on the fields. Its factors are now measurable functions of the local fields and spatial marks, by the preceding input reconstruction. Conditional expectation therefore gives the same product formula given the complete fields and spatial marks. This is exactly the argument of Proposition “Spatial product kernels” in [22], applied to the enlarged output vector of Lemma 2.6.

The time cuts are independent of the continuum background. Area parameterization and the Poisson mapping theorem give their spatial conditional intensity $\rho\mu_h$ for a layer of time intensity ρ . Write Π_i for the marks in the i th domain and $K_i^{\text{marks}}(h, \hat{h}, \Pi_i; \cdot)$ for the conditional output kernel already obtained. Given the fields, the marks have independent Poisson laws, and hence the output law after removing them is

$$\bigotimes_i \int K_i^{\text{marks}}(h, \hat{h}, \pi_i; \cdot) \text{Pois}_{\rho\mu_h|U_i}(d\pi_i).$$

This proves the fields-only assertion. It uses no independence between the marks and the numerical arrays, whose tests may depend on those marks. Alternatively, retain each Π_i together with its array as the local output. Finite joint consistency and the countable prescription library then give the kernel on all tests. Lemmas 2.4 and 2.5 identify its buffered exhaustion with the corresponding observations from the actual graph. No additional conditioning on exterior numerical outputs is used. $\square$

Proposition 2.8 (Coordinate and unit transport). *The kernels in Proposition 2.7 can be retained jointly with the following transformations.*

1. For a fixed conformal coordinate map ϕ , pull back the fields by

$$g \mapsto g \circ \phi + Q \log |\phi'|, \quad \widehat{g} \mapsto \widehat{g} \circ \phi - \chi \arg \phi'.$$

The numerical-array law is transported by the corresponding spatial relabeling of its tests.

2. Fix $v = 2^k$, $k \in \mathbb{Z}$, and put $c_v = \gamma^{-1} \log v$. Expressing the surface in the field $g - c_v$ replaces area time by time divided by v and replaces rate m by rate mv . Electrical energies and extremal lengths are unchanged by this relabeling. If a numerical output is divided by b_m , retain this same factor b_m in the mv experiment of the joint extraction.
3. Write $K_{(1)}$ for the original local law, including the mark kernel, and $K_{(v)}$ for the law of the corresponding transformed rate experiment. Then

$$K_{(v)}(g, \widehat{g}) = \text{Relabel}_v K_{(1)}(g + c_v, \widehat{g}). \quad (2.14)$$

All fixed dyadic indices, their transformed templates, and the required Poisson layers may be retained in one joint extraction.

Proof. This is the finite-experiment identity underlying Proposition “Intrinsic transport and fixed quantum units” in [22]. A conformal change leaves quantum areas, frontier lengths, directed seam roles, and their relative heights unchanged. Hence it leaves the finite instruction graph unchanged and only changes the projection of its labels.

Adding c_v multiplies areas by v and boundary lengths by $\sqrt{v}$. Thus the block inputs transform as

$$\ell \mapsto v\ell, \quad z(t) \mapsto \sqrt{v} z(t/v), \quad o \mapsto \sqrt{v} o. \quad (2.15)$$

Cuts transform with time; band endpoints, their anchors, and selected minima transform with height. A rate- m Poisson string of duration $v\ell$ is, after this time change, a rate- mv string of duration ℓ . The height lattices agree because $\sqrt{v} (mv)^{-1/2} = m^{-1/2}$. Matching strips and minimum comparisons are exactly the same integer tests. Therefore the numerical costs on the resulting graph are identical. Smooth sampling densities may change, but Lemma 2.6 leaves the conditional kernels unchanged.

The cuts must be transformed too. At field $g + c_v$, the layer corresponding to a fixed layer at field g has time intensity divided by v . Its spatial intensity is then unchanged. Choose a countable family of layers closed under these changes, with independent layers and nested refinements as required. Their joint input and output identities give (2.14). A fixed field translation is locally equivalent to the Gaussian reference law, as specified below; this makes the identity valid under that fixed reference marginal as well. No estimate uniform in unbounded v is asserted or needed. $\square$

For exhaustion of the plane we retain the joint cone normalization, not only the deterministic-coordinate kernel identity. In the circle-average embedding define

$$\begin{aligned} r_v &= \sup\{r > 0 : h_r(0) + Q \log r = c_v\}, \\ h^{(v)} &= h(r_v \cdot) + Q \log r_v - c_v, \quad \widehat{h}^{(v)} = \widehat{h}(r_v \cdot), \\ \eta^{(v)}(t) &= r_v^{-1} \eta(vt), \quad Z^{(v)}(t) = v^{-1/2} Z(vt). \end{aligned} \tag{2.16}$$

The complete marginal has the original cone peanosphere law, by the joint zoom in [22] and [5, Proposition 4.13(i)]. The positive dilation preserves the independent stationary-phase imaginary-field law. These transformations are applied to the contour, raw arrays, and full background together, before disintegration. Thus field-dependent reembedding does not require substituting a random map into separately chosen versions of deterministic-coordinate kernels. Moreover $r_{2^k} \rightarrow \infty$ as $k \rightarrow \infty$: the level c_{2^k} diverges, whereas $h_r(0) + Q \log r$ is bounded above on compact sets of radii and tends to $-\infty$ as $r \downarrow 0$. Ordinary patches inside these normalized disks therefore exhaust the plane away from the cone point.

Lemma 2.9 (Ordinary Gaussian reference patches). *On a buffered patch compactly inside the cone’s normalization disk and away from its marked point, the actual real and imaginary fields are locally equivalent to Gaussian reference restrictions. The real-field pinning circle can be fixed outside all testing buffers. This comparison preserves the actual additive constant and can be made after any fixed affine change of coordinates.*

Adjoining any of the local kernels above preserves the same field-only Radon–Nikodym derivative. Consequently one fixed density truncation transfers high-probability events uniformly over all kernels attached to that reference marginal. Fixed additive real-field shifts and the fixed coordinate transforms used in Proposition 2.8 preserve the corresponding local null sets.

Proof. This is Lemma “Ordinary cone patches and exterior Gaussian pins” in [22]. Inside the unit disk the circle-average cone field has the law of the circle-average-zero whole-plane GFF minus $\gamma \log |z|$. On the patch, this deterministic term has a smooth extension supported away from the marked point and the pinning circle. The Cameron–Martin theorem and Gaussian domain comparison give local equivalence, with the actual constant retained. The same argument applies to a fixed additive shift on the patch. The imaginary field is independent and is interpreted modulo its angular period; its local restrictions have the corresponding Gaussian equivalence and stationary-phase coordinate invariance.

If $Z = d\nu/d\mu$ is a field density and K_i any of the conditional kernels, then for every event A and every $R > 0$,

$$(\nu \otimes K_i)(A) \leq R(\mu \otimes K_i)(A) + \mathbb{E}_\mu[Z \mathbf{1}_{\{Z > R\}}]. \tag{2.17}$$

The last term does not depend on i . This proves the uniform probability comparison; local equivalence gives both directions for probability-one events. A test domain containing the full pinning circle is excluded by the stated buffer condition. $\square$

Lemma 2.10 (Primal–dual reflection). *The joint local-array laws are invariant under exchanging the primal and dual graphs and reflecting their coordinate tests. They are also covariant under fixed rigid motions, with the natural transformations of the two fields.*

Proof. The discrete assertion follows by exchanging the two independent coordinate strings and their forests. In the continuum, spatial reflection combined with negation of the reflected imaginary field exchanges the two angle trees and the two sides of the curve. The space-filling

ordering is preserved: it can be read either from the $+\pi/2$ tree with right merges or from the $-\pi/2$ tree with left merges, as in [18, Section 1.2.3, Figure 1.16]. Both descriptions are exchanged by this operation, so no reversal of area time is involved. The cone field and the independent stationary-phase imaginary field have the required reflected law. Boundary lengths exchange L and R , and conditional sampling of the background therefore gives the same operation on the joint raw-array limit. The intrinsic kernels and the selector construction then transport the spatial tests. Fixed rotations and translations follow from coordinate covariance and local field equivalence on buffered ordinary patches. A countable group of such transformations can be retained simultaneously. $\square$

Proposition 2.11 (Local-array interface). *From every sequence of resolutions tending to infinity one can extract, jointly with the continuum peanosphere background, all compactified numerical tests described above, on both graphs and at every fixed dyadic area offset. Prescribed deterministic normalizing factors may be retained even when they tend to zero. In this extraction:*

1. *complete buffered graph pieces and the selector sandwich of Lemmas 2.4 and 2.5 identify local tests with actual graph observations;*
2. *Proposition 2.7 supplies buffer-local conditional kernels and product laws in disjoint domains;*
3. *Proposition 2.8 and Lemmas 2.9–2.10 supply their coordinate, unit, Gaussian-reference, and primal–dual comparisons;*
4. *these properties persist under further joint extraction and under countable exhaustion with strict margins.*

Proof. All numerical coordinates belong to compact spaces, and the input contours are tight on bounded windows. Include the countable transformed instruction lists and mark layers before the diagonal extraction. The preceding results apply to every finite collection. Their exact density and bridge calculation passes to any further limit by compact truncation; the continuum background passes by the L^1 conditional sampling argument. Countable intersection and the selector exhaustions give the stated simultaneous version. $\square$

The interface provides laws and exact local observability. The bounds which prevent degeneration of the electrical tests, and the identification of their variational limits, are proved next.

3 Nondegeneration of the electrical networks

We first prove that a fixed conformal annulus admits cutoffs of bounded electrical energy on both sewn graphs. The bound will depend on the annulus aspect and the coordinate patch, but not on its radius or the discrete resolution. This is the coercivity input for the variational argument in the next section. Throughout this section we work with the bilateral sewing, at rates $m \rightarrow \infty$, and use the coupled extractions and projections π_m of Proposition 2.11.

The proof combines two uses of extremal length. A lower bound supplies densities that charge every crossing of an annulus. If lower bounds failed, a suitable quantile would instead supply probability measures on crossing paths with very small squared traffic. Annular densities make their projected limits rectifiable. Locality then forces a transverse crossing witness on the other graph, contradicting the small traffic. The annular-density, rectifiable-flow, and local-germ strategy is developed for flag extremal length in [21, Section “The canonical conformal structure”]. We adapt it to edge energies on paired primal and dual graphs.

3.1 Extremal length and local germs

Let $\mathcal{P}$ be a nonempty family of paths in a finite graph, with no path of length zero. Edge occurrences are counted with multiplicity. Its edge extremal length is

$$\text{EL}(\mathcal{P}) = \sup_{a:E \rightarrow [0,\infty)} \frac{(\inf_{P \in \mathcal{P}} \sum_{e \in P} a(e))^2}{\sum_{e \in E} a(e)^2}, \quad (3.1)$$

where the zero density is omitted. For a probability measure ν on paths, write $j_\nu(e) = \int \#\{\text{occurrences of } e \text{ in } P\} \nu(dP)$ for its unsigned edge traffic.

Lemma 3.1 (Densities and traffic). *For a finite family of simple paths,*

$$\text{EL}(\mathcal{P}) = \min_{\nu} \sum_e j_\nu(e)^2. \quad (3.2)$$

If $\mathcal{P}$ consists of all paths between two disjoint wired sets in a finite subgraph, then $\text{EL}(\mathcal{P})$ is the reciprocal of the minimum energy $\mathfrak{e}(u)$ subject to the values zero and one on the two sets. In particular, $\text{EL}(\mathcal{P}) \geq b > 0$ supplies a density charging each path by one with squared norm at most $1/b$.

Proof. The traffic vectors form the convex hull of the path-incidence vectors. Let j be its point nearest to the origin. The supporting inequality $\langle j, j_P \rangle \geq \|j\|_2^2$ for every path P shows that the density j gives a value at least $\|j\|_2^2$ in (3.1). Conversely, averaging path length under any ν and applying Cauchy–Schwarz gives $\text{EL}(\mathcal{P}) \leq \|j_\nu\|_2^2$. This proves (3.2). In the wired-set case loops may be removed from paths. Absolute potential increments give a charging density, while the distance from the first set in any charging density, clipped at one, gives an admissible potential whose energy is at most the squared density norm. These two constructions prove the reciprocal identity. $\square$

We will also use a consequence of the local conditional kernels from Section 2. A *point-germ readout* is a jointly measurable random function $Y(z)$ whose value at z can be read from the fields and extracted arrays in every neighborhood of z . The readout may use countably many tests with strictly positive domain margins. In particular, an assertion that specified tests succeed at every sufficiently small scale is a point-germ readout.

Lemma 3.2 (Determinism of point germs). *In an ordinary field chart, every bounded real point-germ readout of the extracted local arrays is, almost surely, equal at Lebesgue-almost every point to a deterministic measurable function of the point. The same conclusion holds for readouts with values in a separable metric space, using a countable family of bounded coordinates that separates its points.*

Proof. Let $\mathcal{H}$ denote the pair of fields. By Proposition 2.7, $\mathbb{E}[Y(z) \mid \mathcal{H}]$ is measurable with respect to the field restrictions to every neighborhood of z . This field germ is trivial. Indeed, for a zero-boundary Gaussian free field, the intersection of the Gaussian subspaces generated by restrictions to shrinking disks is orthogonal to the Gaussian variables associated with Dirichlet functions vanishing near z . Such functions are dense in the Dirichlet space: logarithmic cutoffs express the zero capacity of a point. The intersection is thus zero, and the Gaussian chaos decomposition gives triviality of the corresponding sigma-field. Apply this to both independent fields. Local equivalence in Lemma 2.9 transfers the conclusion to the ordinary chart, including the imaginary field modulo its period. Consequently $\mathbb{E}[Y(z) \mid \mathcal{H}] = y(z)$ for a deterministic y .

For distinct z, w , disjoint sufficiently small neighborhoods and the product-kernel assertion give conditional independence of $Y(z)$ and $Y(w)$ given $\mathcal{H}$. Hence, for every rational disk D compactly in the chart,

$$\mathbb{E} \left[\left(\int_D (Y(z) - y(z)) dz \right)^2 \right] = 0.$$

The diagonal in $D \times D$ has Lebesgue measure zero. Countably many disks and differentiation of integrals prove the assertion. Applying the argument to bounded separating coordinates proves the final statement. $\square$

Remark 3.3 (Coordinate and dual symmetries). For a readout defined intrinsically from local tests, its deterministic values respect the coordinate transports of Proposition 2.8. More explicitly, let $\mathcal{A}$ be the extracted local array and let $T\mathcal{A}$ denote its relabeling by a fixed translation or rotation T . A covariant readout satisfies

$$Y_{T\mathcal{A}}(Tz) = T \cdot Y_{\mathcal{A}}(z),$$

where $T \cdot$ is the natural action on its values or direction labels. The transport proposition identifies the law of $T\mathcal{A}$ with the same extracted kernel attached to the transformed fields. Local equivalence of these field laws and Lemma 3.2 therefore give $y(Tz) = T \cdot y(z)$ almost everywhere. This uses a relabeling in the joint extraction, rather than a newly chosen subsequential kernel. For translation-covariant scalar readouts it implies constancy: first use rational translations, then continuity of translations in local L^1 . Rotations act on direction or matrix readouts in the stated natural way, and a countable dense rotation group suffices. Finally, Lemma 2.10 exchanges the two graphs and reflects the tests. These observations apply on overlapping ordinary charts and identify their deterministic values. The definition through strict margins is important here: a transported test is still an admissible test after an arbitrarily small reduction of its clearance, by Lemma 2.5.

3.2 Annular barriers at absolute quantum scales

Choose once and for all an ordinary cone patch, strictly inside the normalization disk and away from the cone point. After a fixed affine coordinate change, it contains the annulus $\{1/16 < |z| < 4\}$ and the buffers needed below. For each graph take paths between the vertex sets selected by visits in the bands

$$A_1 = \{.9 < |z| < 1.1\}, \quad A_2 = \{1.9 < |z| < 2.1\},$$

confined to the actual subgraph selected in $A_0 = \{.7 < |z| < 2.3\}$. Denote the extremal lengths of these primal and dual path families by Z_m^p and Z_m^d . Here selection uses the continuum visit labels in the coupling, with a fixed step-endpoint convention. This defines a comparison variable; it need not itself be a local instruction test. Properness makes the selected labels finite. All their incidences and adjoining star layers are protected in larger bounded contour windows. The projection and tube properties of Proposition 2.2 therefore apply. If the ports overlap or there is no crossing, give the comparison the value one. The probability of these exceptions tends to zero.

Let $Z_m = \min\{Z_m^p, Z_m^d\}$. Fix $\varepsilon > 0$, to be chosen below, and choose deterministic $q_m > 0$ such that

$$\mathbb{P}[Z_m \leq 2q_m] \geq \varepsilon/2, \quad \mathbb{P}[Z_m \geq q_m/2] \geq 1 - 2\varepsilon. \quad (3.3)$$

We use rates in a countable set closed under dyadic multiplication and division, for example the positive dyadic rationals. The quantiles can be chosen bounded below on every bounded rate interval in $[1, \infty)$. To see this, first bound the continuum preimage window with high probability, then bound the Poisson step count and the degrees of the finitely many encountered vertices. Local finiteness of the bilateral sewing makes these bounds uniform at bounded rates. A nonexceptional finite graph with at most L edges has positive crossing extremal length at least $1/L$.

The comparison in (3.3) produces genuinely local barrier events. For example, select ports near radii .96 and 2.04, and confinement between .94 and 2.06, with strictly positive slack. Every path from $\{|z| < .6\}$ to $\{|z| > 2.5\}$ contains such a crossing portion. A complete local prescription covering this intermediate shell exists by Lemma 2.4; Lemma 2.5 selects the ports and confinement. Its path family is a subfamily of the comparison family, so its EL is no smaller.

The selection can depend on the limiting fields without changing this probability comparison. To see the quantifiers, let j enumerate pairs of primal and dual prescription templates. Let A_j be the local event that both templates' seam, cap, and positive-margin conditions cover the required shell and implement the selector sandwich. These conditions are read from the local fields and marks; their proved consequence is eventual completeness on the actual graph. They do not test completeness by inspecting exterior numerical arrays. For any deterministic units $b_m > 0$ retained in the extraction, write ξ_j for the minimum of the two retained limits of EL/b_m . Thus success supplies barriers on both graphs. If a comparison event asserts $Z_m/b_m \geq a > 0$, retain its indicator in a further coupling. On the event E where the limiting indicator is one, choose an applicable j from the limiting local cell data, keeping it fixed as $m \rightarrow \infty$. Its eventual domination of the comparison gives $\xi_j \geq a$. Consequently

$$E \subseteq \bigcup_j (A_j \cap \{\xi_j > a/4\}).$$

The union on the right is a local event with at least the limiting comparison probability. Simultaneous convergence of the countably retained tests justifies the random choice of j ; no deterministic bound on that index is required. This argument uses the comparison for its probability and the local union for its conditional kernel, without conditioning that kernel on E .

We recall precisely the annulus input that spreads these barriers through a chart. It is Lemma “All-point absolute quantum-scale annuli” in [22, Section “Annuli at absolute quantum scales”]. Write $v_i = 2^{-i}$ for the backward quantum-area offsets. By Proposition 2.8, graph costs are unchanged at these offsets; any deterministic denominator b_m is kept the same across the offset experiments in one extraction.

Lemma 3.4 (Aligned annular barriers). *Fix an ordinary chart and a compact subset of its interior. Suppose that, at every fixed backward index i , a local unit-annulus event has probability at least $1 - \varepsilon_*$ in the reference law and supplies, on both graphs, a crossing density of charge one and squared norm at most C/b_m . Here $b_m > 0$ is deterministic and the events belong to the common extracted offset kernels. For a sufficiently small reference threshold $\varepsilon_* > 0$, there are constants $0 < a < B$, $c > 0$ such that the following holds almost surely for all sufficiently large integers N . Every point in the compact set belongs to the central holes of at least cN pairwise disjoint buffered annuli whose local barrier events hold. Their radii belong to one fixed geometric sequence in $[e^{-BN}, e^{-aN}]$, their centers lie on deterministic meshes of spacing comparable to their radii, and their assigned indices belong to $\{N, \dots, 2N\}$. For each fixed such finite configuration, its density conclusions hold on the discrete graphs for all sufficiently large m .*

Proof. Apply the cited lemma to the kernels of Proposition 2.7, with the intrinsic transport of Proposition 2.8. Its hypotheses are the local product rule, the actual-field coordinate rule, and a common fixed-index change of quantum unit; all are supplied by Proposition 2.11. The real-field parameter $Q = 2/\gamma + \gamma/2$ is greater than two. The cost-index factor in the stated annulus lemma is undone in our numerical tests; the proof of that lemma uses the factor only in the indexed observation rule. Thus it applies equally to the present unchanged network costs. Local Poisson marks are included in the kernels with their corresponding intensities.

For completeness, the probability transfer needed here is a fixed one. Work first with the zero-boundary Gaussian reference patch of that lemma. Local equivalence, retaining the actual additive constant of the real field, gives a single absolute-continuity modulus from our chosen ordinary cone patch to this reference patch. Choose ε in (3.3) sufficiently small that a barrier event of probability $1 - 2\varepsilon$, with a strict threshold as above, has reference probability at least $1 - \varepsilon_*$. The annulus lemma gives an exponentially small failure probability at scale N . Borel–Cantelli gives the eventual assertion. A fixed local change of field law transfers this probability-one assertion to any ordinary chart. No likelihood bound uniform over those charts is needed. Finally, success of an annulus is the existence of a retained EL coordinate strictly above a fixed threshold. No density or path is retained as an additional output. All centers, radii, indices, and prescription templates belong to countable libraries. On their simultaneous convergence event, the successful local prescription on each realized annulus gives an eventual discrete EL lower bound. Lemma 3.1 then constructs a density on that finite graph. For fixed N only finitely many annuli are used, so their thresholds in m can be met simultaneously, even though the successful annuli and prescriptions were selected from the fields and arrays. Thus N is fixed before m is sent to infinity. $\square$

The geometric content of these many annuli is an area bound for a density that charges macroscopic displacement. For a set $U \subset \mathbb{C}$ write $U^{+r} = \{z : \text{dist}(z, U) < r\}$, and put $R_N = e^{-aN}$.

Lemma 3.5 (Densities controlling displacement). *Under the hypotheses of Lemma 3.4, let U be a finite union of open boxes compactly in a fixed ordinary chart. For all sufficiently large N , and then all sufficiently large m , there is on either graph a nonnegative density $g_{m,N,U}$, supported near U , such that*

$$b_m \sum_e g_{m,N,U}(e)^2 \leq C \text{vol}(U^{+CR_N}). \quad (3.4)$$

Every path subinterval whose projection is contained in U pays at least

$$c \text{diam}(\pi_m(P)) - CR_N - o_m(1). \quad (3.5)$$

The constants depend only on the smaller chart and the fixed annular geometry. The error and the threshold in m may depend on N and U . Charges add over disjoint path subintervals.

Proof. On a successful shell of radius r , multiply its barrier density by r . Its charge is now r and its squared norm is at most Cr^2/b_m . Sum these densities over the shells whose central holes meet U , using one density for each geometric shell, and divide by N . At any fixed radius the buffered shells have bounded overlap, because their centers lie on a mesh of comparable spacing. There are at most CN radii. Cauchy–Schwarz across these radii gives (3.4); all the involved shells lie within distance CR_N of U .

For the charge estimate choose a coordinate in which the projected path has oscillation comparable to its diameter. Consider levels between its extreme values, discarding CR_N near

either end. At a visit to each remaining level, the point lies in at least cN central holes. Since all these shells have radius at most R_N , the path crosses each of them before reaching a suitable extreme. A shell of radius r can account for at most Cr of the levels. Integrating over levels therefore bounds the total length in the summed densities from below by cN times the remaining coordinate oscillation. Divide by N . At fixed N the vanishing projected mesh permits full-edge crossing portions; the discarded endpoint pieces contribute $o_m(1)$. Distributing each edge's density uniformly along that edge permits fractional first and last edges and makes additivity over disjoint intervals explicit. $\square$

3.3 Small traffic would force transverse crossings

We now prove the quantitative fact needed for unnormalized energies.

Proposition 3.6 (Nondegeneration). *For the quantiles in (3.3), with ε chosen as in Lemma 3.4,*

$$\liminf_{m \rightarrow \infty} q_m > 0. \quad (3.6)$$

The conclusion holds along all rates in the chosen countable set.

Proof. Suppose otherwise. Choose approximate record lows $m \rightarrow \infty$ such that $b_m := q_m \rightarrow 0$ and

$$q_s \geq cb_m \quad (1 \leq s \leq m, s \text{ in the chosen rate set})$$

for a fixed $c > 0$. The bounded-rate lower bounds ensure $m \rightarrow \infty$. For each fixed i , the quantile at $m2^{-i}$ therefore gives the high-probability barrier event in units b_m . Extract jointly with all these backward offsets. Lemmas 3.4 and 3.5 apply in the comparison region.

The first inequality of (3.3) and Lemma 3.1, on the other hand, supply with positive probability one graph's probability flow ν_m between the two macroscopically separated ports, confined in a fixed compact set, with

$$\sum_e j_{\nu_m}(e)^2 \leq Cb_m. \quad (3.7)$$

Retain the event indicator, choose a graph on a positive-probability subevent, and work pathwise in a further coupling. We establish two properties of the projected flows before using locality.

Compactness of the flows. For a path P , let $J_h(P)$ be the maximum number of disjoint subintervals making a projected advance of at least h . Choose N large enough that $CR_N < ch/2$ in (3.5), and then take m large. Use a fixed larger box union U containing the compact confinement. By (3.4), (3.7), and Cauchy–Schwarz,

$$\int J_h(P) \nu_m(dP) \leq C/h. \quad (3.8)$$

Thus the paths are tight after reparameterization. One explicit choice of parameters is useful. At every dyadic $h = 2^{-l}$ place clock atoms at successive completed moves of distance h , each with mass $2^{-l}/(1 + J_h(P))$. Add the original traversal time of total mass one, invert this clock with pauses at its atoms, and rescale its total duration to one. On sets where every $J_{2^{-l}}$ has a fixed bound, these parameterized paths are equicontinuous: an oscillation of several times 2^{-l} must cross a whole atom interval of a fixed positive duration. The bounds (3.8) and a countable union bound make such sets have arbitrarily large flow probability. The finitely many initial resolutions omitted at an individual scale can be included by enlarging its bound. A subsequence therefore converges to a probability law ν on continuous paths. Their endpoints remain separated.

An *absolutely continuous measure of arclength*. For at most J disjoint path intervals compactly confined in an open box union U , the same argument gives

$$\int \sup \sum_{k=1}^J |\pi_m(P(t_k)) - \pi_m(P(s_k))| \nu_m(dP) \leq C \sqrt{\text{vol}(U + CR_N)} + CJR_N + o_m(1). \quad (3.9)$$

The supremum may be taken over rational intervals with closure in the prescribed open condition. It is lower semicontinuous under uniform convergence. Pass first to the path limit, next let $N \rightarrow \infty$, and finally let $J \rightarrow \infty$. These interval sums exhaust variation on open sets. With a larger U containing all paths, they first show that ν -almost every path is rectifiable with integrable length. Its average arclength measure Λ consequently satisfies

$$0 < \Lambda(\mathbb{C}) < \infty, \quad \Lambda(U) \leq C \sqrt{\text{vol}(U)}. \quad (3.10)$$

The positivity follows from the separated endpoints. Open box covers now show $\Lambda \ll \text{vol}$.

We next convert a tangent of a limiting path into a local event. Choose a sufficiently fine finite grid $\mathcal{W}$ of unoriented directions, closed under quarter-turn and reflection in a coordinate axis. For $z \in \mathbb{C}$, $w \in \mathcal{W}$, and $s > 0$, consider the tube

$$T(z, w, s) = \{z + tw + lw^\perp : |t| < 2s, |l| < s/16\}. \quad (3.11)$$

A *strict crossing test* in this tube has ports strictly beyond the two lines $t = -s$ and $t = s$, and has confinement and ports with positive clearance from the tube boundary. It is computed on a complete local prescription. Accept all countably many selectors with these properties, and require that the prescription be readable within a fixed-factor enlargement of the tube, with positive margins. Lemma 2.5 makes existence of such a test independent of the particular selector library after arbitrarily small relaxation of the margins.

At Λ -almost every point there is a direction $w \in \mathcal{W}$ for which, at every sufficiently small dyadic scale s , one of these tests on the graph supplying ν_m has finite limiting EL $/b_m$. Indeed, arclength parameterization gives a unit tangent at almost every parameter of almost every rectifiable path. The fine direction grid puts that tangent inside one of the tubes with strict slack at all sufficiently small scales. Almost every path is in the support of ν . The existence of the strict crossing subpath is an open path event, so it has positive ν -probability at each such fixed scale. Restrict ν_m to this event, take the crossing portions, and normalize. Its traffic still has squared norm $O(b_m)$, with a constant allowed to depend on the scale and the event probability. The complete-selector sandwich gives the asserted finite limiting test value. The statement can be imposed simultaneously at the countably many dyadic scales.

Let $I_w^p(z)$, respectively $I_w^d(z)$, be the indicator of existence of these finite-EL $/b_m$ tests at every sufficiently small dyadic scale, on the primal or dual graph. These are jointly measurable point-germ readouts: the tests and their fixed-factor buffers eventually lie in every neighborhood of z . The preceding paragraph and (3.10) give a positive-area set of witnesses on our positive-probability event. Lemma 3.2 and Remark 3.3 therefore imply that one of these indicators has deterministic value one almost everywhere in ordinary charts. Reflection exchanges the graphs; rotations within the finite direction grid then give the other graph's indicator in the perpendicular direction. The null sets can be discarded simultaneously, since only finitely many directions and two graphs are involved.

Choose a point with both perpendicular indicators equal to one, and then a sufficiently small common dyadic scale. We obtain primal and dual strict crossing families with EL $= O(b_m)$ in

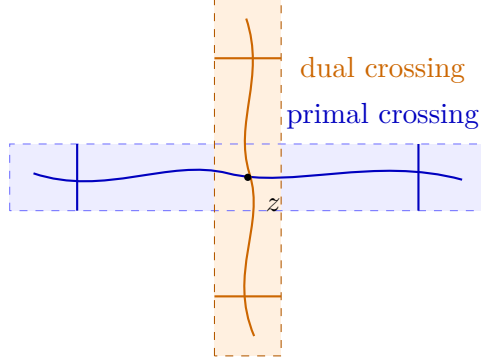


Figure 2: The two narrow tubes force transverse passages. The drawing shows only projected path ranges; it asserts neither injectivity of the projection nor shape control of discrete edges. The topological input lifts this configuration to an intersection of a primal edge with its paired dual edge.

perpendicular tubes, as illustrated in Figure 2. Every path of one family meets every path of the other upstairs by the transverse-tube rule in Proposition 2.2. Such an intersection uses an edge and its dual. Choose traffic minimizers j_m, j_m^* by Lemma 3.1. Their unsigned pairing satisfies

$$1 \leq \sum_e j_m(e) j_m^*(e^*) \leq \left(\sum_e j_m(e)^2 \right)^{1/2} \left(\sum_{e^*} j_m^*(e^*)^2 \right)^{1/2} = O(b_m),$$

contradicting $b_m \rightarrow 0$. This proves (3.6). $\square$

3.4 Uniformly bounded ring cutoffs

We can now apply the annular construction in constant energy units, rather than the hypothetical vanishing units used in the contradiction.

Proposition 3.7 (Ring cutoffs). *Fix a compact subpatch of an ordinary chart and an aspect ratio $\tau > 1$. Along a coupled further extraction from any rate sequence, there is a constant C , depending only on the chart and τ , with the following property. For every fixed disk $\overline{B(z, \tau r)}$ contained with a margin in that subpatch, both graphs eventually admit functions χ_m valued in $[0, 1]$ such that*

$$\chi_m = 1 \quad \text{on every lift of } \overline{B(z, r)}, \quad \chi_m = 0 \quad \text{on every lift outside } B(z, \tau r), \quad \mathfrak{e}(\chi_m) \leq C$$

on the primal graph, with $\mathfrak{e}^(\chi_m) \leq C$ on the dual. Their energy is supported in the indicated ring, up to vanishing projected mesh. The constant is independent of z, r, m ; the threshold in m may depend on the fixed disk. These assertions hold simultaneously for the countable buffered disk tests, and extend to arbitrary fixed disks by strict inner and outer margins.*

Proof. Proposition 3.6 and the second quantile bound in (3.3) give the barrier hypotheses with deterministic units equal to one, at every fixed backward offset. Choose an intermediate closed ring whose two radii lie strictly between r and τr , with all three resulting radial widths comparable to $(\tau - 1)r$. Cover it by a finite union U of open boxes with closure strictly inside the target ring and area $O_\tau(r^2)$. Apply Lemma 3.5 to this U . First choose N so large that R_N is small

compared with the intermediate width and both collars, and then take m large. The density is supported in the target ring. Every full ring crossing has a subcrossing through the intermediate ring, confined to U , with advance comparable to $(\tau - 1)r$. The density therefore charges it by a fixed multiple of this width. After division by the width, the squared density norm is bounded by $C(\tau)$ times the chart constant. The clipped-distance construction of Lemma 3.1 gives the cutoff, with exact constants on all inner and outer lifts. One may carry out the construction on a protected capped sphere and extend by constants beyond the chart. Only the ring density changes the function.

Countable buffered comparisons make all the fixed tests available in one extraction. A disk outside the chosen countable library is handled by choosing a nearby library center, an inner library disk containing $\overline{B(z, r)}$, and a concentric outer library disk contained in $B(z, \tau r)$. Their radii can be chosen comparable to r , with constants depending only on τ , and their radial separation at least $c(\tau - 1)r$. The preceding density estimate bounds the cutoff energy by a chart constant times the squared ratio of outer radius to radial separation. It therefore gives a uniform $C(\tau)$ for these library tests, with the required constants on the original inner and outer sets. This use does not require a resolution threshold uniform over all disk radii. $\square$

All these conclusions are local in the actual conformal coordinate. They therefore also hold in each fixed forward quantum-area offset of Proposition 2.8. The joint cone zoom and ordinary-patch exhaustion of [22, Lemma “From indexed Gaussian charts to the whole cone”] let these patches exhaust the plane away from its marked cone point. Only probability-one conclusions are transferred in this exhaustion. The marked point will be handled by removability when the variational limits are assembled.

4 The limiting electrical energy

We now identify the unnormalized energies of the primal and dual graphs. The cutoffs of Proposition 3.7 first give compactness for harmonic functions and coercivity for arbitrary continuous limits. The locality of the extracted arrays then makes the limiting energy a deterministic scalar multiple of the conformal Dirichlet integral. An annular current–period argument will show that this scalar is exactly one. Thus the normalization is obtained from the network itself.

Throughout this section we work in an ordinary conformal chart, in a coupled extraction with the conclusions of Proposition 2.11 and Proposition 3.7. Write G_m, G_m^* for the primal and dual graphs and π_m for their surface projections. All sets used below have fixed positive clearance inside the chart before m tends to infinity. The energy $e_m(v; U)$ is the sum of $|v(y) - v(x)|^2$ over edges with both projected endpoints in U ; we use the same convention for compact sets. Replacing this convention by edge-location membership does not change an inner limit, because the projected mesh tends to zero. Uniform convergence on a compact set always means convergence at every vertex over that set, including every lift of a point.

4.1 Circuits and harmonic compactness

A cutoff on one graph gives a short level circuit on the other, where “short” is measured by the increments of the function under study. This elementary coarea observation is the bridge between the cutoffs and the analysis of functions.

Lemma 4.1 (A separating circuit). *Fix concentric disks with radii $0 < r < R$ and with a neighborhood of the closed larger disk inside an ordinary chart. Given a primal function v_m , there is a simple primal circuit C_m in a slightly buffered annulus which surrounds all lifts of the smaller disk and whose inside contains no lift outside a slightly enlarged larger disk, such that*

$$\left(\sum_{e \in C_m} |dv_m(e)| \right)^2 \leq C \mathbf{e}_m(v_m; A^+). \quad (4.1)$$

Here A^+ is a fixed buffered annulus and C depends only on its aspect and the chart constants. The conclusion holds eventually for every fixed annulus. It also holds with the primal and dual graphs exchanged.

Proof. Take a dual cutoff b_m which is one on the inner well and zero on the outer well, with energy at most C . These wells include all relevant lifts, with fixed collars. For $t \in (0, 1)$ away from the finitely many vertex values, color the primal faces according to whether their dual vertices satisfy $b_m > t$. The oriented boundary of the colored faces splits into simple circuits. A face appearing on both sides of an edge contributes no boundary occurrence; other edge occurrences have their usual multiplicities.

The boundary chain has winding difference one between an inner dual vertex and an outer dual vertex, because their face colors are one and zero. After splitting the chain into simple circuits, at least one circuit separates these two vertices. The inner dual well is connected through dual edges on which the cutoff is one; such a path cannot cross any boundary edge, including at a primal articulation vertex. Hence every vertex of this inner well is on the same side of the chosen circuit. The connected outer well has the analogous property. The collars and vanishing overlay mesh put all primal inner lifts on the inner side as well, and all lifts beyond the larger well on the outer side. This uses the all-lift joining statement of Proposition 2.2 and remains valid when the full boundary chain touches itself. The selected circuit lies in the change annulus and has variation at most that of the entire boundary. The coarea identity and Cauchy–Schwarz give

$$\int_0^1 \sum_{e \in \partial\{b_m > t\}} |dv_m(e)| dt = \sum_e |dv_m(e)| |db_m(e^*)| \leq \mathbf{e}_m(v_m; A^+)^{1/2} \mathbf{e}_m^*(b_m)^{1/2}.$$

Choose a level at most this average. This proves (4.1), with an arbitrarily small slack if necessary. The same argument applies after exchanging the graphs. $\square$

Lemma 4.2 (Harmonic compactness). *Let U be a connected open subset of an ordinary chart. Suppose that v_m is graph-harmonic at every vertex whose star projects into U , and that its energies are bounded on compactly contained neighborhoods. Then v_m is macroscopically equicontinuous on every compact subset of U : for $K \Subset U$,*

$$\lim_{\delta \downarrow 0} \limsup_{m \rightarrow \infty} \sup_{\substack{\pi_m x, \pi_m y \in K \\ |\pi_m x - \pi_m y| < \delta}} |v_m(x) - v_m(y)| = 0.$$

If x_m is a vertex sequence with $\pi_m x_m \rightarrow z_0 \in U$ and $v_m(x_m)$ is bounded, the functions have locally uniformly convergent subsequences. The assertions hold for both graphs and for local branches of potentials, with a normalization on each initial branch.

Proof. Fix a disk with clearance inside U and an energy bound E on a larger disk. Around its center place J disjoint annuli of one fixed aspect, all inside a prescribed small neighborhood. Choose all radii before taking m large. Their energy sum is at most E , so Lemma 4.1 supplies a surrounding circuit on which the oscillation is at most $(CE/J)^{1/2}$. The discrete maximum principle bounds the oscillation everywhere inside by the same quantity. Taking J large, and then using finitely many such neighborhoods, proves equicontinuity. No assertion is needed at a radius depending on m .

The same construction at a fixed, nonvanishing scale bounds oscillation on compact balls. Overlapping chains of balls propagate an additive normalization throughout a connected compact subset. Vertex density and the small-image joining property identify the limits on overlaps. The usual finite-net proof of Arzelà–Ascoli now gives local uniform compactness. A local potential branch is an ordinary harmonic function on each of its branch disks, so the proof also applies to branches. $\square$

4.2 The uniform relaxation and its energy measures

The extracted variational tests let us define a single limiting energy for every continuous target. This definition does not assume a continuum Dirichlet form.

For $u \in C(U)$, $K \Subset U$ compact, and $\delta > 0$, let

$$M_m(u, K, \delta) = \inf \{ \mathbf{e}_m(v; K) : |v(x) - u(\pi_m x)| \leq \delta \text{ for every vertex } x \text{ over } K \}.$$

Define the *inner relaxed energy* by

$$F(u, U) = \sup_{K \Subset U} \sup_{\delta > 0} \liminf_{m \rightarrow \infty} M_m(u, K, \delta). \quad (4.2)$$

One may use a countable exhaustion of compact finite unions of boxes and let δ decrease through positive rationals. Define F^* by the same formula on the dual graph.

Lemma 4.3 (Uniform variational limits). *The value in (4.2) is unchanged if $\liminf$ is replaced by $\limsup$. It is lower semicontinuous for locally uniform convergence and inner regular in U . Moreover:*

- (i) *Every locally uniformly convergent sequence $v_m \rightarrow u$ has local energy lower bounds given by $F(u, \cdot)$, also along further subsequences.*
- (ii) *If $F(u, U) < \infty$, there are compact sets K_m exhausting U and functions v_m on their graph neighborhoods such that $v_m \rightarrow u$ locally uniformly and $\mathbf{e}_m(v_m; K_m) \leq F(u, U) + o(1)$.*

These statements hold simultaneously for all continuous targets. The values $F(u, U)$ are locally readable from the arrays with arbitrarily small open buffers.

Proof. Given $K \Subset K' \Subset U$ and $0 < \delta' < \delta$, insert between the two tests a finite selector test from Lemma 2.5. Cover the relevant visit labels by finitely many selectors on which the target varies by less than a fixed fraction of $\delta - \delta'$, and use intermediate rational value constraints. Completeness of the selected graph preserves every edge incidence. Restriction of an admissible function for the larger test is admissible for the intermediate one; restriction from the intermediate test is admissible for the smaller one. The intermediate infimum has a retained limit. Consequently,

$$\limsup_m M_m(u, K, \delta) \leq \liminf_m M_m(u, K', \delta').$$

Taking the two inner suprema proves the assertion about $\liminf$ and $\limsup$, including infinite values.

The same comparison proves the lower bound in (i). For (ii), choose almost minimizers on increasing compact sets with decreasing tolerances, and take a diagonal sequence using the $\limsup$ formula. Tolerance slack also proves lower semicontinuity, directly from (4.2). Exhausting U proves inner regularity. The selector library is countable; approximation of a continuous target uses its modulus of continuity on each compact, rather than a new exceptional event. This gives the simultaneous assertion. The locality and measurability claims follow from the same selector sandwich, with its spatial buffers. $\square$

We call the sequences in Lemma 4.3(ii) recovery sequences. The next estimate explains why such sequences can be joined on overlaps. For an oriented edge xy , put $\bar{v}(xy) = (v(x) + v(y))/2$.

Lemma 4.4 (Gluing estimate). *Let v_m, w_m converge uniformly to the same continuous function on a fixed transition region. Suppose $0 \leq b_m \leq 1$ has bounded energy, is supported where v_m is defined, and equals one wherever w_m is not used. Then $z_m = b_m v_m + (1 - b_m) w_m$ has, on the gluing region,*

$$\begin{aligned} \mathbf{e}_m(z_m) &\leq (1 + \varepsilon) \sum_e \left(\bar{b}_m(e) |dv_m(e)|^2 + (1 - \bar{b}_m(e)) |dw_m(e)|^2 \right) \\ &\quad + (1 + \varepsilon^{-1}) \|v_m - w_m\|_{\infty, \text{tr}}^2 \mathbf{e}_m(b_m), \quad \varepsilon > 0. \end{aligned} \quad (4.3)$$

Finite partitions of unity have the corresponding estimate, with an error tending to zero when the local approximations agree uniformly on overlaps. Bounded-energy gluing weights exist for every fixed finite open cover of a compact set.

Proof. The exact difference identity is

$$dz_m = \bar{b}_m dv_m + (1 - \bar{b}_m) dw_m + \overline{v_m - w_m} db_m.$$

Apply convexity of the square to the first two terms, followed by $|a + c|^2 \leq (1 + \varepsilon)|a|^2 + (1 + \varepsilon^{-1})|c|^2$. Only the transition region contributes to the last term. For finitely many weights the same identity follows by summing and using $\sum_i db_{i,m} = 0$.

To construct weights, choose finitely many disks whose smaller disks cover the compact set and whose larger disks stay in the prescribed members of the cover. Use Proposition 3.7, sum the resulting cutoffs, and normalize where their sum is at least one. Clipping and division on this fixed range preserve bounded energy. Extra cutoffs with collars extend the construction to a neighborhood of the compact set. All these choices precede the limit in m . $\square$

Proposition 4.5 (Quadratic energy measures). *For each open U , the functional $F(\cdot, U)$ is quadratic on its finite-energy space and vanishes on constants. If u has locally finite relaxed energy, then*

$$U \mapsto F(u, U)$$

is the restriction to open sets of a Radon measure. For smooth u , on a smaller ordinary chart,

$$F(u, U) \leq C \int_U |\nabla u|^2 dz. \quad (4.4)$$

The same statements hold for F^ .*

Proof. Recovery sequences for u and v , the discrete parallelogram identity, and the energy lower bound give

$$F(u + v, U) + F(u - v, U) \leq 2F(u, U) + 2F(v, U).$$

Applying this inequality to $u + v$ and $u - v$ gives the reverse inequality. Homogeneity follows directly from the tolerance definition. Constant addition leaves every edge increment unchanged.

For fixed u , restriction gives monotonicity and superadditivity on disjoint open sets. To prove subadditivity on $U \cup V$, first fix a compact subset of this union. Recover u in U and V , and join the recoveries with Lemma 4.4; the transition is compactly contained in $U \cap V$. Let $m \rightarrow \infty$ and then $\varepsilon \downarrow 0$ in (4.3). Exhausting the compact gives

$$F(u, U \cup V) \leq F(u, U) + F(u, V).$$

Finite subcovers and inner regularity imply countable subadditivity. For completeness, these properties give the required measure criterion as follows. Define $\nu(A) = \inf\{F(u, O) : A \subset O, O \text{ open}\}$. This is an outer measure by countable subadditivity. If two sets have positive distance, they have disjoint open neighborhoods; disjoint-open superadditivity shows that ν is additive on such sets. Thus it is a metric outer measure and all Borel sets are measurable. On an open set O , monotonicity gives $\nu(O) = F(u, O)$. Local finiteness and inner regularity make this Borel measure Radon.

It remains to prove (4.4); this will also establish local finiteness for smooth targets. On a compact set inside U , use a square grid of spacing r and a bounded-overlap family of disk cutoffs with radii comparable to r . Normalize these cutoffs to weights $b_{z,m}$ which sum to one near the compact, and set

$$v_m = \sum_z b_{z,m} u(z).$$

For fixed r this approximates u within its modulus of continuity at scale Cr . On each edge the weight increments sum to zero. Subtracting one nearby node value and using bounded overlap bounds the energy by

$$C \sum_z r^2 \sup_{B(z, Cr)} |\nabla u|^2.$$

Indeed each unnormalized cutoff has bounded energy, and normalization has the same bound because the denominator is bounded below and only boundedly many supports meet an edge. First take m large for this finite collection, and then let $r \downarrow 0$. The displayed sum converges to at most a constant times the gradient integral over the chosen region. Exhaustion inside U proves (4.4). The arguments are unchanged on the dual. $\square$

4.3 Integral representation and coercivity

The energy measures of the two coordinate functions determine the whole functional. Coercivity is a separate step: it follows from intersecting level circuits and does not assume geometric regularity of the mesh.

Proposition 4.6 (Integral representation). *In every ordinary chart there is a measurable symmetric matrix field A , locally bounded and uniformly positive definite on smaller charts, such that, for every open U and every $u \in C(U)$,*

$$F(u, U) = \begin{cases} \int_U \nabla u^\top A \nabla u \, dz, & u \in W_{\text{loc}}^{1,2}(U), \\ +\infty, & u \notin W_{\text{loc}}^{1,2}(U). \end{cases} \quad (4.5)$$

The integral is allowed to be infinite. There is an analogous field A^* for the dual functional.

Proof. We first identify the energy of smooth functions. Polarize the Radon measures from Proposition 4.5. The measures of the linear functions $\ell_1(z) = \Re z$, $\ell_2(z) = \Im z$ and their sums are absolutely continuous by (4.4). Define the entries of A by their densities:

$$A_{ii} dz = F(\ell_i, dz), \quad 2A_{12} dz = F(\ell_1 + \ell_2, dz) - F(\ell_1, dz) - F(\ell_2, dz).$$

Quadraticity and the upper bound imply that A is symmetric, nonnegative, and locally bounded. At a Lebesgue point z of these densities, compare a smooth u with its affine first-order Taylor polynomial p_z . The seminorm inequality and (4.4) give

$$\left| F(u, B(z, r))^{1/2} - F(p_z, B(z, r))^{1/2} \right| \leq C^{1/2} |B(z, r)|^{1/2} \sup_{B(z, r)} |\nabla u - \nabla u(z)|.$$

Divide by $|B(z, r)|^{1/2}$ and let $r \downarrow 0$. Differentiation of Radon measures yields (4.5) for smooth u .

We next prove the converse bound, including weak differentiability:

$$\int_U |\nabla u|^2 dz \leq CF(u, U). \quad (4.6)$$

Suppose the right side is finite. Put a square grid of fixed small spacing r in a compactly contained region of U . At every grid node x , apply Lemma 4.1 to a recovery sequence, using an annulus with inner and outer radii $0.7r$ and $0.8r$, and choose a value $a_{x,m}$ at a vertex on the resulting circuit. Buffers are small fixed fractions of r .

Circuits at neighboring grid nodes share a vertex. Here is the necessary topological verification. A point near the midpoint of the two centers is inside both inner disks. Points on the two opposite sides belong to exactly one inner disk and lie beyond the other outer disk; a distant point lies outside both. The all-lift separation rule therefore gives a common interior lift, a common exterior lift, and lifts with each of the two opposite memberships. Two disjoint simple circuits on the sphere with these memberships could be neither nested nor disjoint in their interiors, a contradiction. Since both circuits belong to the primal graph, their intersection contains a vertex. Consequently,

$$|a_{x,m} - a_{y,m}|^2 \leq 2 \left(\sum_{e \in C_{x,m}} |dv_m(e)| \right)^2 + 2 \left(\sum_{e \in C_{y,m}} |dv_m(e)| \right)^2$$

for neighbors x, y . Write B_r for the union of the buffered grid annuli. Bounded overlap and (4.1) give

$$\sum_{x \sim y} |a_{x,m} - a_{y,m}|^2 \leq C e_m(v_m; B_r) \leq C(F(u, U) + o(1)).$$

For the last inequality, this fixed finite union lies in the recovery set K_m for all sufficiently large m . The piecewise affine interpolation of the $a_{x,m}$ on grid triangles has Dirichlet energy bounded by this sum times an absolute constant. Take m large after fixing the grid, and then let $r \downarrow 0$. Continuity of u and uniform recovery imply that these interpolations converge locally uniformly to u . Weak Sobolev compactness proves (4.6) on the compact region; exhaustion proves it on U . Applying it to smooth affine functions on arbitrary disks gives $A \geq cI$ locally.

Finally, let $u \in C(U) \cap W_{\text{loc}}^{1,2}(U)$. On smaller regions, smoothing approximates u both uniformly and in $W^{1,2}$. Lower semicontinuity and the smooth upper bound first extend that

upper bound to such continuous Sobolev functions. Apply the extended bound to the difference between u and its smooth approximations. Their F -seminorm difference tends to zero, so the smooth formula passes to u . Exhaustion gives (4.5), including infinite integrals. If local Sobolev regularity fails, (4.6) forces infinite energy. All steps apply equally to F^* . $\square$

Lemma 4.7 (The deterministic scalar). *In each coupled extraction there is a deterministic $\lambda \in (0, \infty)$ such that*

$$A = A^* = \lambda I \quad (4.7)$$

almost everywhere in all ordinary charts. At this stage λ may depend on the extracted rate sequence.

Proof. The derivatives of the energy measures of linear functions are point germs of the arrays: use the tolerance definition on successively smaller disks, with arbitrarily small open buffers. Choose measurable lim sup versions and a fixed default on the null set where Lebesgue differentiation fails. Lemma 3.2 applies to these matrix-valued readouts (or to a bounded injective encoding of their entries), so A and A^* are deterministic almost everywhere. Translation covariance makes each field a constant matrix on overlapping ordinary charts. Under a rotation, the matrix transforms by $A \mapsto RAR^T$. Rotation covariance, in particular a quarter-turn, therefore makes each symmetric matrix scalar. These uses of covariance are precisely those in Remark 3.3, and compare local laws with their actual field transformations. Primal–dual reflection (Lemma 2.10) equates the two scalar values. The upper bound and coercivity give $0 < \lambda < \infty$. $\square$

4.4 Annular reciprocity fixes the scalar

We have proved existence and isotropy of the limiting energy. To identify its normalization, compare a condenser potential with a potential whose value increases by one on going around an annulus. The first comparison rotates a discrete current to a dual potential; the reverse comparison recovers an angular potential with its period *exactly* preserved. Both comparisons use only interior graph pieces with collars. The current–potential correspondence is the standard planar-network duality; see [17, Section 9.2]. Its use to determine a scalar effective conductivity has a classical precedent in Dykhne’s planar self-duality argument [6]. Here we give both inequalities, including the recovery needed on our irregular graph pieces.

Let

$$A(r, R) = \{z : r < |z| < R\}, \quad C_0 = \frac{2\pi}{\log(R/r)},$$

where the closed annulus and a neighborhood of it lie inside an ordinary chart. Thus C_0 is the ordinary Dirichlet capacity between its two boundary circles. The radial minimizer is $\log(|z|/r)/\log(R/r)$. A locally Sobolev potential with additive period P around the annulus has energy at least

$$\int_{A(r,R)} |\nabla v|^2 dz \geq \frac{P^2}{C_0}. \quad (4.8)$$

Indeed, on almost every circle, $\int_0^{2\pi} \partial_\theta v d\theta = P$; apply Cauchy–Schwarz and integrate ds/s . Equality holds for $v = P\theta/(2\pi)$, interpreted in local branches.

Lemma 4.8 (Condenser convergence). *Prescribe $u_m = 0$ on every primal vertex over $|z| \leq r$ and $u_m = 1$ on every vertex over $|z| \geq R$, using a protected cap outside the chart. Let C_m be the minimum primal energy. Then*

$$C_m \longrightarrow \lambda C_0. \quad (4.9)$$

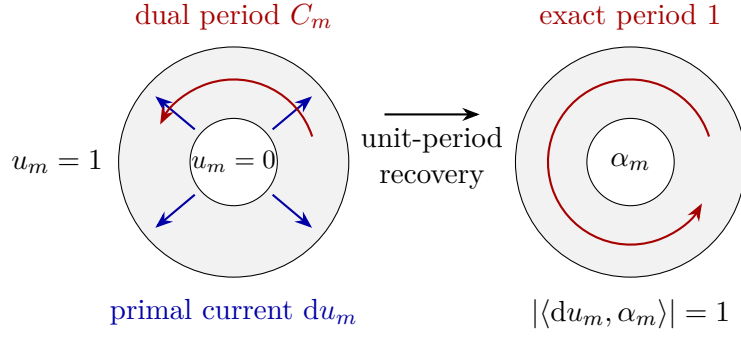


Figure 3: The two annular comparisons. Rotating the condenser current gives a dual potential with period C_m and hence $\lambda \leq 1$. An exact unit-period dual recovery pairs with the unit condenser jump to give $\lambda \geq 1$. This is a schematic of the test annulus; the projected discrete graph need not resemble a regular mesh.

The minimizers converge uniformly, including at the two plate boundaries, to the round radial condenser potential.

Proof. All nonzero increments of the minimizer project into any fixed collar of the closed annulus eventually. Indeed, both endpoints of an edge strictly on one plate have its prescribed value, and the overlay mesh vanishes. Thus the cap contributes no uncontrolled energy term.

For the upper bound, approximate the radial potential by smooth functions which are constant on slightly wider plate collars and whose energies tend to C_0 . Recover them uniformly on a neighborhood of the annulus. Bounded-energy weights in the flat collars paste in the exact constants at a vanishing energy cost, by Lemma 4.4. Lemma 4.7 gives $\limsup C_m \leq \lambda C_0$.

The minimizer lies in $[0, 1]$ and is harmonic off the plates. Its bounded energy and Lemma 4.2 give compactness in the open annulus. At a plate-boundary point, use the stacks of rings from that lemma, small enough to avoid the other plate. Every surrounding circuit meets a path lying strictly on the plate side: choose a tube there from the inner well to outside the ring and lift it by Proposition 2.2. Thus the circuit contains a vertex with the prescribed value. A circuit of small variation is uniformly close to that value. Inside it all remaining vertices are harmonic or have that same prescribed value, so the maximum principle proves continuity uniformly up to the plate. Fixed stacks followed by finite covers give the assertion simultaneously along both boundaries.

Any uniform subsequential limit has the condenser boundary values. The variational lower bound and Lemma 4.7 give energy at least λC_0 . This proves (4.9); equality in the round Dirichlet principle identifies the limit and gives the last assertion. $\square$

Lemma 4.9 (Rotation of a current). *Rotate the primal current du_m from Lemma 4.8 to the dual edges, with one consistent crossing orientation. In the annulus this is a closed dual cochain with harmonic local potential branches and period $\pm C_m$. The branches have subsequential locally uniform limits with period $\pm \lambda C_0$ and with total dual relaxed energy at most λC_0 over one turn.*

Proof. Closedness around a dual face is the primal harmonic equation. Harmonicity of a potential branch at a dual vertex is the telescoping sum of du_m around the corresponding primal face. These identities count all incidences and are valid also for loops and repeated edges. The squared dual increments sum to at most C_m on every interior annular piece.

We justify the existence of branches without assuming regular discrete annular boundaries. In a disk with clearance inside the annulus, arbitrary dual lifts in a smaller disk are connected within the larger one. Loops used to integrate the cochain can be filled inside a further buffer, by Proposition 2.2; every primal vertex there is harmonic. Cellular closedness therefore makes the integral independent of the chosen path. On connected overlaps the resulting potentials differ by constants.

A dual loop with projected winding one can be built along fixed tubes and closed using small-image joining. The integer-winding statement of Proposition 2.2 makes its winding upstairs one, up to the fixed orientation sign. Its current integral is the total current entering one plate, since all intervening primal vertices are harmonic. Discrete summation by parts gives total current C_m : the potential difference is one and its energy is C_m . Hence the period is $\pm C_m$.

Normalize branches on a finite overlapping disk cover of any compact subannulus, propagating constants through chosen overlaps. In particular, fix a cyclic chain of disks around one positively oriented circle. Continue a normalized branch along this chain, matching on each connected overlap; on returning to the first disk, the continued branch differs from the original by $\pm C_m$. Lemma 4.2 gives local uniform subsequential limits. Their transition constants converge, so their period is $\pm \lambda C_0$. Exhausting the annulus yields branches everywhere. For the energy bound, use disjoint boxes lying in branch disks. Their discrete edge energies do not overlap, and their sum is at most C_m . Apply the local lower bound and then exhaust the annulus up to sets of Lebesgue measure zero. This gives the asserted bound over one turn. $\square$

Combining Lemmas 4.7 and 4.9 with (4.8) gives

$$\lambda C_0 \geq \lambda \frac{(\lambda C_0)^2}{C_0}, \quad \text{hence } \lambda \leq 1. \quad (4.10)$$

For the reverse inequality, angular recovery must retain its period before taking any limit. We record first the measure observation that makes the gluing sharp.

Lemma 4.10 (Energy measures of recovery sequences). *Let V be a bounded patch and let p be smooth on a neighborhood of its closure. Choose a recovery sequence on compact sets exhausting V whose total energy tends to $F^*(p, V)$. Count only edges with both endpoints in these recovery sets, and put their squared increments at any of the corresponding projected edge locations. These measures converge vaguely in V to $\lambda |\nabla p|^2 dz$. In particular, energies weighted by a compactly supported continuous function converge to the corresponding weighted integral.*

Proof. The total masses are bounded. Let ν be a subsequential vague limit. To check its local lower bound, take open sets $W \Subset O \Subset V$ and a compact set $K \subset O$ containing W with a collar. The energy tests inside W are eventually counted by the edge measure $\nu_m(K)$. Lemma 4.3 and compact-set Portmanteau give

$$F^*(p, W) \leq \liminf_m \nu_m(K) \leq \limsup_m \nu_m(K) \leq \nu(K) \leq \nu(O).$$

Exhaust O by such W . Thus ν dominates $\lambda |\nabla p|^2 dz$. Its total mass is at most the limiting total recovery energy, which is the total mass of that same measure. Equality follows. Every subsequential vague limit is consequently the stated measure, proving the claim. $\square$

Lemma 4.11 (Recovery with exact unit period). *For every sufficiently small $\eta > 0$, let $A_\eta = A(r - \eta, R + \eta)$. On the dual edges in a neighborhood of the closed annulus there are closed cochains α_m with exact period one and*

$$\limsup_{m \rightarrow \infty} \sum_{e^* \subset A_\eta} |\alpha_m(e^*)|^2 \leq \lambda \left(\frac{1}{C_0} + o_\eta(1) \right), \quad o_\eta(1) \rightarrow 0. \quad (4.11)$$

The cochains are closed at every cell whose star lies in this neighborhood; the region can be chosen to contain every edge where $du_m \neq 0$ for all large m .

Proof. Take finitely many disks covering a slightly larger closed annulus, each avoiding the origin, with angle branches $p_i = \theta_i/(2\pi)$. Recover p_i uniformly on larger branch disks. Lemma 4.10 applies to these recoveries, denoted $v_{i,m}$. Take a smooth partition of unity b_i with supports compactly inside the branch disks. Recover its weights, clip them to be nonnegative, cut them off in fixed support collars, and normalize their sum. This gives $b_{i,m} \rightarrow b_i$ uniformly, bounded energies, supports inside the branch disks, and $\sum_i b_{i,m} = 1$ near the annulus.

There is a useful canonical cochain even though its own energy need not be bounded. On an oriented dual edge, let

$$a_m(e^*) = \frac{1}{2\pi} \int_{\pi_m(e^*)} d \arg z.$$

Here the integral means the endpoint increment of a continuous argument lift along the projected edge. Every sufficiently small edge image lies in a branch disk, so this is well defined. Vanishing face mesh makes a_m closed around every cell under consideration, and its period around a loop is that loop's projected integer winding.

On the support of $b_{i,m}$ the error $v_{i,m} - p_i \circ \pi_m$ is a single-valued function tending uniformly to zero. Define the globally single-valued error and cochain by

$$q_m = \sum_i b_{i,m}(v_{i,m} - p_i \circ \pi_m), \quad \alpha_m = a_m + dq_m. \quad (4.12)$$

The support collars make this definition independent of how terms are extended by zero. Since a gradient has zero periods, α_m is closed and has period exactly one, in the projected orientation. This conclusion is an identity at each finite resolution.

To bound its energy, apply the discrete product identity on each edge. On a branch disk, $d(p_i \circ \pi_m) = a_m$. Since the weights sum to one, the terms containing a_m cancel to give

$$\alpha_m(e^*) = \sum_i \bar{b}_{i,m}(e^*) dv_{i,m}(e^*) + \sum_i \overline{v_{i,m} - p_i \circ \pi_m}(e^*) db_{i,m}(e^*).$$

The second sum tends to zero in edge ℓ^2 , by uniform convergence of the errors and bounded energy of the finitely many weights. Convexity bounds the squared norm of the first sum by the weighted sum of local recovery energies. Lemma 4.10 and $b_{i,m} \rightarrow b_i$ turn its limit into

$$\lambda \sum_i \int_{A_\eta} b_i |\nabla p_i|^2 dz = \lambda \int_{A_\eta} \left| \nabla \frac{\theta}{2\pi} \right|^2 dz = \frac{\lambda}{2\pi} \log \frac{R + \eta}{r - \eta}.$$

One may compute first on a slightly larger closed annulus inside the cover; its boundary circles have zero limiting energy measure. The edge averages $\bar{b}_{i,m}$ converge uniformly to b_i at the chosen edge locations, by vanishing mesh. This justifies both the weighted passage to the limit and restriction to the annulus. This proves (4.11). Finally, u_m is constant on each plate, so its nonzero increments lie in A_η eventually by vanishing edge mesh. Choosing additional fixed collars gives the assertion about stars. $\square$

Proposition 4.12 (Exact conductance normalization). *The scalar in Lemma 4.7 equals one.*

Proof. Only $\lambda \geq 1$ remains after (4.10). Pair du_m with the cochain α_m of Lemma 4.11, matching each primal edge with its crossing dual edge and using the consistent crossing sign. Then

$$\left| \sum_e du_m(e) \alpha_m(e^*) \right| = 1. \quad (4.13)$$

To see this at the discrete level, replace u_m by the indicator of the outer side of an intermediate circle. Their difference is supported strictly inside the region of closedness of α_m . Cellular summation by parts therefore leaves the pairing unchanged. The indicator's gradient pairs with α_m along the oriented dual boundary chain separating the two wells. This chain has total winding one up to sign by Proposition 2.2; the cochain has exact unit period by (4.12). This proves (4.13). The argument uses the boundary chain with its multiplicities and does not require a simple discrete plate boundary.

Cauchy–Schwarz, (4.9), and (4.11) now give

$$1 \leq C_m \sum_{e^* \subset A_\eta} |\alpha_m(e^*)|^2 \implies 1 \leq \lambda C_0 \lambda \left(C_0^{-1} + o_\eta(1) \right).$$

Let $\eta \downarrow 0$. Thus $1 \leq \lambda^2$, and (4.10) proves $\lambda = 1$. $\square$

Theorem 4.13 (Uniform conformal energy limit). *In every coupled extraction and every ordinary chart, both unnormalized network energies have the same inner relaxed limit:*

$$F(u, U) = F^*(u, U) = \int_U |\nabla u|^2 dz \quad (4.14)$$

for every $u \in C(U) \cap W_{\text{loc}}^{1,2}(U)$, with infinite value when the integral diverges or local Sobolev regularity fails. The local uniform lower bounds and recovery conclusions of Lemma 4.3, the gluing estimate, and harmonic compactness hold simultaneously. In particular, the limiting energy has no subsequence-dependent multiplicative constant.

Proof. Combine Proposition 4.6, Lemma 4.7, and Proposition 4.12. The preceding construction used the same countable buffered tests throughout, so its simultaneous and locality assertions persist. All arguments apply in every fixed quantum-unit offset of the extraction. Their compatibility is the transport statement of Proposition 2.11. $\square$

5 From local plane energies to the sphere

The energy identification of Section 4 was obtained in ordinary patches of a quantum cone. We now transfer it to the fixed-area sphere and include the point at which the contour begins and ends. The output is a global variational statement in the same coupling as the metric-measure convergence. This common coupling will later allow us to compare the walks and their metric spaces simultaneously.

Write dA for round area on the conformal sphere, and set

$$\mathcal{D}(u) = \int_{S^2} |\nabla u|^2 dA \quad \text{for } u \in C(S^2) \cap W^{1,2}(S^2),$$

with $\mathcal{D}(u) = +\infty$ for other continuous functions. The same notation on an open coordinate domain means the integral over that domain, with local weak differentiability required. Conformal invariance makes these definitions independent of the chosen chart.

5.1 Transfer on the interior of a contour interval

First extend the plane conclusion beyond a single ordinary patch. Retain all forward dyadic unit changes $v = 2^k$ from Proposition 2.8. For each fixed k , the transformed rate still tends to infinity, so Theorem 4.13 applies on the ordinary patches inside that cone's normalization disk. Under the joint zoom (2.16), these disks exhaust the plane away from the cone point. Pulling the countably many probability-one conclusions back to the original coordinate proves the local energy identity everywhere off that point. Both the network costs and the two-dimensional Dirichlet integral are unchanged by these transformations.

For $0 < d < 1/2$, let $I_d = [d, 1 - d]$. The decorated interval surface associated with the contour on I_d has an interior determined by those increments. A compact subset of that interior, together with a small collar, has all its contour visits strictly inside I_d . Thus its eventual discrete incidences are read from the internal word; no information from either omitted end of the tour is needed. These are the interval-surface and complete-core assertions used in Lemma 2.4.

Here is the part of the excursion comparison that permits new variational observations. It concerns the joint law of the arrays and the word, not independence after weak convergence.

Lemma 5.1 (Additional observations under excursion conditioning). *Fix $0 < a < b < 1$. Let W_N be the diffusively rescaled free axial increment path on $[[aN], [bN]]$, with linear interpolation, and let Z_N be any measurable function of these increments with values in a fixed compact metrizable space. Suppose $(W_N, Z_N) \Rightarrow (W, Z)$ along a sequence of even N . There is a nonnegative continuous function $f_{a,b}$ of the limiting path, with expectation one, such that the corresponding pair under quadrant-excursion conditioning converges to the law*

$$f_{a,b}(w) \text{ Law}(W, Z)(dw, dz).$$

In particular, every probability-one property of (W, Z) also holds for this conditioned limit. A countable product of compact-valued observations is allowed for Z_N .

Proof. The lemma *Excursion densities* in the spanning-tree sphere section of [22] proves the following statements for the density f_N of this increment block. If $w_N \rightarrow w$ uniformly, then $f_N(w_N) \rightarrow f_{a,b}(w)$; the limit depends continuously on the two coordinate minima and the net increment. Moreover, $\sup_N \mathbb{E}[f_N(W_N)^2] < \infty$ under the free law. These assertions follow there from the entrance-and-exit killed-walk density and its Gaussian bounds.

Realize the assumed joint convergence almost surely. For any bounded continuous test Φ , first truncate the densities at height K . Sequential convergence and bounded convergence give convergence of the weighted expectations with truncated densities. The error in removing truncation is at most $K^{-1} \|\Phi\|_\infty \sup_N \mathbb{E}[f_N(W_N)^2]$, and the analogous limit error has the same bound. Hence

$$\mathbb{E}[f_N(W_N) \Phi(W_N, Z_N)] \longrightarrow \mathbb{E}[f_{a,b}(W) \Phi(W, Z)].$$

Taking $\Phi = 1$ gives total mass one. The finite-resolution change of density depends only on the increment block, so its weighted expectation is precisely that of the additional block data under excursion conditioning. This proves the assertion. $\square$

Proposition 5.2 (Energy identification under the sphere conditioning). *Consider any joint subsequential extraction of the sphere contour and the countable internal variational arrays on I_d , for every rational $d \in (0, 1/2)$, using the extraction convention of Proposition 2.11. Almost surely, on every open domain compactly contained in the interior of one of these interval surfaces,*

both primal and dual inner relaxed energies equal $\mathcal{D}$. This identity holds simultaneously for all continuous targets, with the value $+\infty$ when the target is not locally Sobolev or its Dirichlet integral diverges.

Proof. At each resolution retain the energy minima with rational value constraints and finite unions of relative-time intervals as selectors. Only increments within I_d enter these internal graphs and tests. Their compactified values are therefore permissible additional block data in Lemma 5.1. The corresponding free experiment is the bilateral sewing at rate $m = 2n$. In that experiment adjoin these jump-indexed internal tests to the Poisson-instruction arrays before extraction. Each added test is computed from its exact integer increment block and is unchanged when an independent Poisson clock is attached to the word. The earlier local identity applies in this enlarged extraction. To read it from the internal tests, use complete pieces and fixed spatial collars as described below. The vanishing discrepancy of the rescaled clocks only locates the selectors inside these collars; the numerical energy outputs themselves are retained unchanged, without a continuity claim for them under relabeling.

Starting with the given conditioned extraction, take a further subsequence on which the corresponding free block and its internal array converge jointly. This is possible by tightness of the free contour and compactness of the array space, simultaneously for the countably many intervals by diagonal extraction. The weighted-limit lemma then identifies the already specified conditioned block limit with the density tilt of this free joint limit.

Theorem 4.13 identifies the inner relaxed energies in that bilateral experiment, away from its distinguished point. The distinguished point does not belong to this interval surface: time zero is almost surely a singleton Brownian contour fiber. Indeed, each coordinate takes values strictly below its value at zero arbitrarily close to zero on both sides. Thus no nontrivial identification link in either coordinate can have zero as an endpoint, and neither can the equivalence relation generated by those links identify zero with another time. Since $I_d \subset (0, 1)$, its image excludes the distinguished point.

To express the local energy conclusion in terms of the block alone, reconstruct its abstract decorated interval surface H_d from the limiting increments using the interval reconstruction in Proposition 2.1. This reconstruction supplies the surface and its restricted curve modulo conformal coordinates. Evaluate the local identity in a conformal representative, using a countable atlas and compact exhaustions of its interior; conformal invariance makes the event independent of that representative. Both the restricted curve locating the labels and the coordinate domains for the targets belong to this reconstructed interval surface. For a compact K in one such domain and an open enlargement U with a collar, Lemma 2.4 supplies all incidences near K from the internal word. The selectors of Lemma 2.5 put every constraint on K between constraints on internal graphs contained in U . Rational tolerances and compact exhaustion therefore express the inner relaxed energy using only the limiting block and its retained arrays. A change of the conformal representative preserves both the transported energy identity and the Dirichlet integral. The resulting assertion is intrinsic to H_d ; the surrounding cone or sphere embedding supplies no further data for its definition.

The simultaneous quantifier on targets can also be retained in this transfer. On a compact coordinate set use a countable dense library of continuous functions and rational tolerances. Uniform closeness of two targets enlarges or shrinks a tolerance constraint by at most that distance. The library therefore describes all the relaxed value tests, jointly measurably in the retained arrays and the target. This is a discretization of constraints, not an approximation of Dirichlet energy merely from values on a uniformly dense library. For varying charts use a

countable atlas and compact exhaustions. The usual Dirichlet integral, with its infinite-value convention, is likewise Borel on continuous targets: weak derivatives and their energy can be tested against countably many smooth functions on the exhaustion. Existence of a continuous target violating the identity is consequently an analytic event, obtained by projection from its Polish function space. The event of equality for all targets is universally measurable. Theorem 4.13 gives precisely this simultaneous event under the free law, so absolute continuity applies to it without a target-dependent exceptional set.

Absolute continuity of the joint block law now transfers the probability-one identity to the excursion experiment. A countable intersection over d , charts, and exhaustions proves the result. The rest of the sphere contour may be retained in this coupling: the complement of the event already has zero probability in the internal marginal, hence also in the joint law. $\square$

The interval interiors exhaust the sphere away from the tour endpoint, denoted by p . In fact, the inverse image under the continuous contour of a compact set disjoint from p is compact and avoids times 0 and 1; the same holds after a sufficiently small spatial enlargement. The interval-surface identification then places the set and its collar in an interior with $d > 0$. Consequently Proposition 5.2 gives the local energy identity at every point other than p .

5.2 Removing the tour endpoint

There are two different issues at p . A limit of bounded-energy functions must have no extra Sobolev singularity there, and recovery sequences must incur no extra discrete energy there. The following elementary fact handles both.

Lemma 5.3 (Continuous Sobolev removability). *Let p be a point of a smooth two-dimensional surface. If u is continuous near p , is locally $W^{1,2}$ off p , and has finite Dirichlet integral near p , then u is $W^{1,2}$ across p . Moreover, if u is continuous and Sobolev on a compact surface, there are continuous Sobolev functions u_r , constant on a neighborhood of p and equal to u outside a shrinking disk, such that*

$$\|u_r - u\|_\infty \longrightarrow 0, \quad \|u_r - u\|_{W^{1,2}} \longrightarrow 0.$$

Proof. Work in a coordinate disk centered at p . A point has zero two-dimensional Sobolev capacity: radial logarithmic cutoffs equal to one on a shrinking inner disk and zero outside a fixed disk have Dirichlet energies tending to zero. Testing the weak-derivative identity off p with the complementary cutoffs proves the identity across p . More explicitly, the additional term containing the cutoff gradient tends to zero by Cauchy–Schwarz, since u is bounded on the support of a fixed smooth test function, while the term containing ∇u converges by its local square integrability.

For the approximation choose a smooth cutoff χ_r equal to one on $B(p, r)$, zero outside $B(p, 2r)$, taking values in $[0, 1]$, and with $\int |\nabla \chi_r|^2 dA \leq C$ independently of r . Set

$$u_r = u(p) + (1 - \chi_r)(u - u(p)).$$

The uniform error is bounded by the oscillation of u on $B(p, 2r)$. For the gradient error,

$$\int |\nabla(u_r - u)|^2 dA \leq 2 \int_{B(p, 2r)} |\nabla u|^2 dA + 2C \sup_{B(p, 2r)} |u - u(p)|^2,$$

and both terms tend to zero. The ordinary L^2 error follows from uniform convergence. $\square$

In a coupled extraction, choose a location $\pi_n(v) \in S^2$ for each primal vertex from one of its contour visits; different choices have uniformly vanishing discrepancy. Use π_n^* on the dual. For functions on the varying vertex sets, uniform convergence to u means

$$\max_{v \in V_n} |u_n(v) - u(\pi_n(v))| \longrightarrow 0.$$

The small-image joining and mesh assertions in the input guarantee that this convention agrees with the local comparisons already used. Harmonicity on an open set U means that the graph Laplacian vanishes at every vertex whose location is in U . For a compact subset of U , all adjoining stars also lie in U once n is large; thus this convention permits the buffered uses of Lemma 4.2.

Theorem 5.4 (Global sphere energies). *From any subsequence of the tree-decorated sphere maps, one can take a further joint extraction and a coupling with the limiting ordinary quantum sphere such that the following statements hold almost surely.*

1. *If u_n converges uniformly to $u \in C(S^2)$, then*

$$\liminf_{n \rightarrow \infty} \mathbf{e}_n(u_n) \geq \mathcal{D}(u).$$

2. *For each $u \in C(S^2) \cap W^{1,2}(S^2)$ there are uniformly convergent approximations u_n for which*

$$\mathbf{e}_n(u_n) \longrightarrow \mathcal{D}(u).$$

Both assertions hold on the dual with $\mathbf{e}_n^$ and π_n^* .*

3. *On every open set, graph-harmonic functions with bounded energy and bounded absolute value are locally equicontinuous under these locations and have locally uniformly convergent subsequences. This holds also on the dual and at the tour endpoint.*

All these conclusions may be required in the same extraction as the metric and measure comparison below.

Proof. Retain the internal tests of Proposition 5.2 and also the whole-sphere tests defining uniform relaxed energy. The latter are countable compactified arrays; they need no local conditional kernel. The selector sandwich applies to them in the same way.

Suppose first that $\liminf_n \mathbf{e}_n(u_n) < \infty$ and pass to a subsequence realizing this liminf. On disjoint compact-interior patches away from p , Proposition 5.2 bounds their limiting Dirichlet integrals by the discrete energies in those patches. Exhausting the complement of p gives

$$\int_{S^2 \setminus \{p\}} |\nabla u|^2 dA \leq \liminf_n \mathbf{e}_n(u_n).$$

Lemma 5.3 supplies the weak derivative at p . This proves the first assertion, including its infinite-value case.

For the upper bound, first take a continuous Sobolev target that is constant near p . Cover the complement of a smaller constant disk by finitely many patches from Proposition 5.2. Subdivide it into finitely many smaller coordinate regions with piecewise smooth boundaries, each contained in one patch, and thicken these regions slightly. The boundaries have zero round area, so their collars can be chosen with arbitrarily small total integral of $|\nabla u|^2$. Local recovery on the thickened regions and successive applications of Lemma 4.4 now give recovery on the compact

complement: the main energy is bounded by the sum of the target energies on those regions, and only the chosen small collars are counted more than once. The error terms from gluing tend to zero by uniform agreement on each fixed collar. This proves recovery with arbitrarily small energy slack. Extend by the constant on the smaller disk. To make the extension exact, use a cutoff whose transition lies in a fixed annulus where the target is constant. Apply the finite-cover recovery just proved to a slightly larger annular neighborhood and a smooth cutoff b that is constant on its two collars. If b_n is its recovery and $\delta_n > \|b_n - b\|_\infty$ tends to zero, replace b_n by

$$\left(\frac{b_n - \delta_n}{1 - 2\delta_n}\right) \vee 0 \wedge 1.$$

This is exactly zero and one on the respective flat collars, and its energy is at most $(1 - 2\delta_n)^{-2}e_n(b_n)$. The functions being joined differ by $o(1)$ on the transition, so the extra term in the gluing estimate is $o(1)$. The mesh tends to zero, hence every edge crossing either gluing boundary is contained in the fixed collar for all sufficiently large n .

For a general continuous Sobolev u , apply the approximation u_r of Lemma 5.3. Recover each fixed u_r first, then use a diagonal choice of $r \downarrow 0$. Uniform convergence and convergence of $\mathcal{D}(u_r)$ give the desired upper bound; the first assertion makes it an equality. The proof for the dual is identical.

Away from p , harmonic compactness follows from Lemma 4.2. To include p , take any fixed finite stack of disjoint annuli centered at p , each of the same aspect ratio, in a domain of harmonicity. Every transition annulus avoids p . By the preceding local recovery, its dual cutoff can be taken with energy bounded by a constant depending only on the aspect ratio, with exact constant values on its two sides. Fill the inner side by the constant; this uses only a flat collar in the annulus. Lemma 4.1 gives a primal circuit in each annulus whose oscillation squared is bounded by that constant times the primal energy in the annulus. Among k disjoint annuli, one therefore has oscillation at most $Ck^{-1/2}$ when the total energy is bounded. The maximum principle transfers this oscillation to all vertices inside its circuit. First choose k , and then take large n ; no annulus radius depends on n . Finite covers give equicontinuity, and bounded values give compactness. Exchanging primal and dual proves the remaining assertion. $\square$

5.3 Keeping the metric and the quantum area in the same coupling

The coordinate used for energy convergence must also describe the metric-measure limit. The contour correspondence makes this precise.

Proposition 5.5 (Joint spatial comparison). *The extraction in Theorem 5.4 can be chosen so that its limiting sphere is the one in the metric-measure limit and, almost surely,*

$$\sup_{v, w \in V_n} |a_n d_n(v, w) - D_h(\pi_n(v), \pi_n(w))| \longrightarrow 0, \quad (5.1)$$

$$\sup_{x \in S^2} \text{dist}_{D_h}(x, \pi_n(V_n)) \longrightarrow 0, \quad (5.2)$$

$$(\pi_n)_* \mu_n \Longrightarrow \mu_h. \quad (5.3)$$

The maximum round diameter of a projected edge tends to zero. The measure μ_h is nonatomic and has full support, and D_h induces the round topology.

Proof. The sphere-identification proposition and the final normalization argument for spanning-tree maps in [22] give, jointly with the excursion, the contour correspondence

$$\{(v_{\lfloor 2nt \rfloor}, \eta(t)) : 0 \leq t \leq 1\}$$

with distortion tending to zero. Here v_j is the primal vertex at the start of tour step j and η is the sphere contour. Those results identify the metric of this very peanosphere limit; they do not introduce an independent quantum sphere. Their deterministic diameter normalization is the a_n in Theorem 1.1. Choosing one visit for each vertex changes the correspondence by a uniformly vanishing discrepancy, which gives (5.1) and (5.2).

There are $2n$ tour step starts. Their vertex occupation measure is μ_n : horizontal tree traversals are counted at their starting vertices, and vertical steps count the non-tree half-edges at their incident vertices. Every half-edge is counted once, including both half-edges of a loop. The area parameterization of η therefore gives (5.3). Mesh convergence is the sphere projection construction preceding Proposition 2.2. Nonatomicity, full support, and agreement of the metric and conformal topologies are the quantum-area and metric properties in the same repository input.

All these variables may be retained together with the compactified energy arrays before extraction. The metric identification is unchanged upon a further extraction, so it holds in the coupling of Theorem 5.4. $\square$

6 A Green estimate in the finite map law

Uniform approximation of energies concerns continuous functions. To control the walk clock, we also need to show that bounded sources of small stationary mass have small potentials. We prove this directly under the finite map law. The argument combines a spanning-forest formula for the inverse Laplacian with the two tree contours in Mullin's encoding.

Write $N = 2n$. Let H_n be the positive graph Laplacian on V_n , using counting measure and unit conductances, so that

$$\langle u, H_n u \rangle = \mathbf{e}_n(u) = \sum_{ab \in E(M_n)} (u(a) - u(b))^2.$$

There is one term for each unoriented edge, and loops contribute zero. For a real function f with $\mu_n(f) = 0$, define $G_n f$ by

$$H_n G_n f = \mu_n f, \quad \mu_n(G_n f) = 0. \quad (6.1)$$

Here $\mu_n f$ on the right denotes the vector with entry $\mu_n(v)f(v)$ at v . Connectedness gives existence and uniqueness.

Proposition 6.1 (Small-mass Green estimate). *For each fixed $\rho \in (0, 1/2)$ there are nonnegative random variables A_n , measurable functions of M_n , such that*

$$\sup_{n \geq 1} \mathbb{E} A_n < \infty, \quad \|G_n f\|_\infty \leq A_n \mu_n(|f|)^\rho \quad (6.2)$$

simultaneously for all real functions $f : V_n \rightarrow \mathbb{R}$ satisfying $\|f\|_\infty \leq 1$ and $\mu_n(f) = 0$. The expectation is under the map marginal of the uniform rooted map–spanning-tree pair law. In particular, (A_n) is tight.

We first identify the exact inverse formula. We then estimate its terms by grouping tree cuts according to the amount of tour time on their smaller side.

6.1 The inverse Laplacian as a sum over cuts

The next identity is a consequence of the matrix-tree theorem and its forest expansions; see the all-minors theorem of [4] and [17, Chapter 4] for electrical networks and spanning trees. We give the determinant argument to specify the multiplicities and centering used here.

Lemma 6.2 (Forest representation). *Let $\mathcal{G}$ be a finite connected multigraph, with positive probability measure μ on its vertex set V , and let H be its unit-conductance graph Laplacian. Suppose $q : V \rightarrow \mathbb{R}$ satisfies $\sum_{v \in V} q(v) = 0$, and let u solve $Hu = q$, $\mu(u) = 0$. For a uniform spanning tree T of $\mathcal{G}$ and an edge $e \in T$, choose either component S_e of $T \setminus e$, and let b_e be the number of graph edges between S_e and $V \setminus S_e$, with multiplicity. Then*

$$u(x) = \mathbb{E}_T \sum_{e \in T} \frac{q(S_e)(\mathbf{1}_{\{x \in S_e\}} - \mu(S_e))}{b_e}, \quad x \in V, \quad (6.3)$$

where $q(S) = \sum_{v \in S} q(v)$. The value does not depend on the choice of component for any e .

Proof. The one-vertex case is immediate, so suppose $m := |V| \geq 2$. Orient the non-loop edges arbitrarily and delete the row of a fixed vertex o from the incidence matrix. Denote the resulting matrix by B , and write $\hat{q}$ for q with its o -coordinate deleted. The reduced Laplacian is $\hat{H} = BB^\top$. If τ is the number of spanning trees, the matrix-tree theorem and the determinant lemma give

$$\det(\hat{H} + z\hat{q}\hat{q}^\top) = \tau(1 + z\hat{q}^\top \hat{H}^{-1} \hat{q}). \quad (6.4)$$

For $z \geq 0$, apply Cauchy–Binet to the matrix obtained by appending $\sqrt{z}\hat{q}$ to B . A term contributing to the coefficient of z selects $m - 2$ incidence columns and the source column. The incidence columns are independent exactly when their edges form a spanning forest with two components. Indeed, a cycle gives a linear dependence, whereas an acyclic graph with $m - 2$ edges has exactly two components, counting isolated vertices.

For such a forest F , let S_F be its component not containing o . Its corresponding determinant has absolute value $|q(S_F)|$. To see this, successively eliminate leaf rows and their incident forest columns. Eliminating a leaf adds its source entry to its neighbor's entry. The component containing o disappears into the deleted row, and the last entry in the other component is $\sum_{v \in S_F} q(v)$. Consequently, comparison of the coefficients of z in (6.4) yields

$$\hat{q}^\top \hat{H}^{-1} \hat{q} = \frac{1}{\tau} \sum_F q(S_F)^2. \quad (6.5)$$

Let H^+ be the inverse of H on counting-mean-zero vectors, extended by zero on constants. Solving the equation with value zero at o shows that the left side of (6.5) equals $q^\top H^+ q$. Polarization therefore gives, for any two vectors q, r of total sum zero,

$$q^\top H^+ r = \frac{1}{\tau} \sum_F q(S_F) r(S_F). \quad (6.6)$$

Replacing S_F by its complement changes the sign of each factor, so the product is independent of this choice.

Every two-component forest F has exactly $b(F)$ extensions to a spanning tree: add any one of the graph edges across its cut. Thus, when all pairs (T, e) are counted, $F = T \setminus e$ occurs exactly $b(F)$ times. This explains the denominator in (6.3). Apply (6.6) with $r = \delta_x - \mu$. Its left side is

$$(H^+ q)(x) - \mu(H^+ q) = u(x),$$

and its right side, regrouped by (T, e) , is the asserted expression. Loops enter neither the reduced incidence matrix nor a cut; parallel edges are counted separately throughout. $\square$

6.2 What a tree cut records in the tour

We use the finite form of Mullin’s correspondence [20], with the incidence conventions of [22, Section “Independent-walk encoding and local quantum surfaces”]. The N steps form a quadrant excursion whose horizontal and vertical coordinates are denoted by L and R . Horizontal steps traverse the selected primal tree T ; vertical steps traverse the complementary dual tree T^* . In each coordinate an up-step is matched with the down-step that first returns to its initial height. A matched horizontal pair represents a tree edge. A matched vertical pair represents a non-tree map edge between the primal vertices at its two step starts. Let v_i be the primal vertex at step start i , with all indices read cyclically modulo N .

Lemma 6.3 (Tour mass and cut size). *For each $v \in V_n$,*

$$\#\{0 \leq i < N : v_i = v\} = \deg_{M_n}(v). \quad (6.7)$$

For $e \in T$, the step starts visiting either component of $T \setminus e$ form a cyclic interval. Choose a component S_e whose interval I_e has length $\ell_e \leq N/2$. If the lengths agree, choose the component not containing v_0 . Then

$$\mu_n(S_e) = \frac{\ell_e}{N}, \quad b_e = 1 + D_e^*, \quad (6.8)$$

where D_e^ is the distance in T^* between the dual vertices at the two horizontal traversal steps of e .*

Proof. Associate each step start with the departing step. The two horizontal traversals of a tree edge depart once from each endpoint, and hence count its two incidences. The two steps of a vertical matched pair count the two incidences of its non-tree edge. When that edge is a loop, both are assigned to the same vertex. Every step belongs to one such pair, proving (6.7); this is also the tour-measure lemma of [22, Section “From tree excursions to the measured quantum sphere”].

The primal vertex changes only along horizontal steps. It passes between the two components of $T \setminus e$ exactly at the two traversals of e . Thus each side occupies one cyclic interval, and (6.7) proves the mass identity.

There is exactly one tree edge across this cut, namely e . A non-tree edge crosses it precisely when exactly one step of its vertical matched pair belongs to I_e . Ignore horizontal holds and follow the resulting dual-tree contour segment. An edge of T^* is traversed once on this segment exactly when its matched steps lie on opposite sides of I_e ; every other dual edge is traversed zero or twice. In a tree the edges traversed an odd number of times by a path are exactly the edges on the simple path between its endpoints. These endpoints are the dual vertices at the two horizontal traversals of e , since horizontal steps do not change the dual vertex. There are therefore D_e^* non-tree edges across the cut, giving $b_e = 1 + D_e^*$. $\square$

Hereafter expectations involving both T and the map use the uniform pair law. The sequence (v_i) is determined by the horizontal contour and the locations of the horizontal slots. Thus the cut intervals, their lengths, and the smaller-side convention in Lemma 6.3 are determined without inspecting the shape of the vertical contour. By (6.7), the same is true of the vertex masses.

The conditional structure of the finite law is useful. Let J be the number of horizontal steps. Conditional on J , the horizontal slots form a uniform J -subset of $\{0, \dots, N - 1\}$. Conditional

on these slots, the two coordinate strings, with holds deleted, are independent uniform Dyck excursions of lengths J and $N - J$. In fact, writing $C_k = (k + 1)^{-1} \binom{2k}{k}$ for the Catalan numbers, the number of quadrant excursions with $J = 2k$ is

$$\binom{2n}{2k} C_k C_{n-k}, \quad \sum_{k=0}^n \binom{2n}{2k} C_k C_{n-k} = C_n C_{n+1}. \quad (6.9)$$

The last equality is Vandermonde's identity after writing out the Catalan numbers. These counts also prove the stated conditional independence. Catalan estimates and a binomial tail bound give absolute $C, c > 0$ such that

$$\mathbb{P}\{J \notin [N/4, 3N/4]\} \leq C(N + 1)^3 e^{-cN}. \quad (6.10)$$

The same schedule calculation is recorded in the excursion discussion of [22, Section "From tree excursions to the measured quantum sphere"]. It is an exact calculation in the fixed-size law.

6.3 Distances in a uniform contour tree

The cut denominator in (6.8) is a dual-tree distance. The next estimates quantify its typical size and also control how many primal cuts can surround a given tour time.

Lemma 6.4 (Contour distance estimates). *Let $\mathcal{T}_s$ be a uniform rooted plane tree with $s/2$ edges, where $s \geq 2$ is even, and write w_i , $i \in \mathbb{Z}/s\mathbb{Z}$, for its cyclic contour vertices. If i, j are deterministic with $0 \leq j \leq s$, set*

$$D = \text{dist}_{\mathcal{T}_s}(w_i, w_{i+j}), \quad d = 1 + \min(j, s - j).$$

There are absolute constants $C, c > 0$ such that, for every integer $a \geq 0$,

$$\mathbb{P}(D = a) \leq C(a + 1)^2 d^{-3/2} e^{-ca^2/d}. \quad (6.11)$$

Consequently, for every fixed $p > 0$,

$$\|(1 + D)^{-1}\|_{L^2} \leq C d^{-1/2}, \quad \|D\|_{L^p} \leq C_p \sqrt{d}. \quad (6.12)$$

For every deterministic cyclic segment K from corner i through corner $i + L$, where $1 \leq L \leq s$,

$$\|\text{diam}_{\mathcal{T}_s}\{w_{i+r} : 0 \leq r \leq L\}\|_{L^4} \leq C\sqrt{L}. \quad (6.13)$$

Proof. Rerooting a plane tree at its i th contour corner is a bijection on rooted plane trees with $s/2$ edges. Its inverse reroots by $s - i$ corners. Thus the rerooted tree is uniform, and D has the law of the height at time j of a uniform Dyck excursion of length s .

Let $p_l(a)$ be the probability that a simple symmetric walk started at zero stays nonnegative for l steps and finishes at a . Reflection and the binomial estimate give

$$p_l(a) = 2^{-l} \left[\binom{l}{(l+a)/2} - \binom{l}{(l+a+2)/2} \right], \quad (6.14)$$

$$p_l(a) \leq C(a + 1)(l + 1)^{-3/2} e^{-ca^2/(l+1)},$$

with inadmissible binomial coefficients interpreted as zero. For even s , $p_s(0)$ is comparable to $(s+1)^{-3/2}$. The Markov property and time reversal of the second segment give

$$\mathbb{P}(D = a) = \frac{p_j(a)p_{s-j}(a)}{p_s(0)}.$$

Inserting (6.14) proves (6.11). Summing it after multiplication by $(a+1)^{-2}$ or by a^p proves (6.12). In the first case the remaining Gaussian sum is at most $C\sqrt{d}$, so the second moment is at most C/d .

For completeness, the diameter estimate follows by dyadic chaining. Reroot at the deterministic first corner of K and parametrize its steps by the integers in $[0, L]$. At refinement level l , use the times $\lfloor kL/2^l \rfloor$, $0 \leq k \leq 2^l$, and stop at $l = \lceil \log_2 L \rceil$. These grids are nested and the last one contains every integer in $[0, L]$. Include the initial increment from 0 to L . For $l \geq 1$, there are $O(2^l)$ new-to-parent increments at level l , each spanning at most $CL2^{-l}$ steps. By the fourth-moment estimate and $\|\max_k Z_k\|_{L^4}^4 \leq \sum_k \|Z_k\|_{L^4}^4$, the maximum tree distance across these increments has L^4 norm at most

$$C2^{l/4}\sqrt{L2^{-l}} = C\sqrt{L}2^{-l/4}.$$

Every contour vertex can be joined to the first vertex through its ancestors in the dyadic grids. The triangle inequality and the summability of $2^{-l/4}$ bound the maximum distance from that first vertex by $C\sqrt{L}$ in L^4 . The diameter is at most twice this maximum. For $L = 1$ the bound follows directly from the single edge. $\square$

6.4 The sum of reciprocal cut sizes at one scale

For a dyadic integer M with $1 \leq M \leq N/2$, let

$$\mathcal{E}_M = \{e \in T : M \leq \ell_e < 2M\}, \quad U_{M,t} = \sum_{\substack{e \in \mathcal{E}_M \\ t \in I_e}} \frac{1}{b_e}, \quad t \in \mathbb{Z}/N\mathbb{Z}.$$

The intervals in this sum come from the horizontal tree. Their denominators are governed by the independent vertical tree once the schedule and the horizontal tree have been fixed.

Lemma 6.5 (One-scale estimate). *There is an absolute constant C such that, for every n , every dyadic $M \leq N/2$, and every deterministic tour time t ,*

$$\mathbb{E}U_{M,t}^2 \leq C. \tag{6.15}$$

Proof. First condition on a count $J \in [N/4, 3N/4]$. Consider all cyclic intervals containing t whose lengths lie in $[M, 2M)$ and are at most $N/2$. There are at most $C(M+1)^2$ such intervals. For each one, a uniform schedule puts at least c_0M vertical steps in both the interval and its complement, except with probability Ce^{-cM} , for suitable absolute $c_0, c > 0$. Indeed, the vertical density is at least $1/4$, both sets have at least M slots, and the hypergeometric lower-tail bound applies. Taking a union bound, let $\mathcal{B}_{M,t}$ be the event that this property fails for some interval. Uniformly over the balanced counts,

$$\mathbb{P}(\mathcal{B}_{M,t} \mid J) \leq C(M+1)^2 e^{-cM}. \tag{6.16}$$

Fix a schedule outside $\mathcal{B}_{M,t}$ and then condition on the horizontal Dyck excursion. This fixes all intervals in $U_{M,t}$. For each one, its dual endpoints are separated in the vertical contour by the number j of vertical steps in that interval. Both j and $(N - J) - j$ are at least $c_0 M$. Lemmas 6.3 and 6.4, followed by the triangle inequality in L^2 of the remaining vertical excursion, imply

$$\|U_{M,t}\|_{L^2(R|\text{schedule},L)} \leq \frac{C}{\sqrt{M}} K_{M,t}, \quad K_{M,t} := \#\{e \in \mathcal{E}_M : t \in I_e\}. \quad (6.17)$$

We next bound this number of horizontal cuts. Any two components $S_e, S_{e'}$ containing v_t are nested unless their union is all of V_n . This elementary tree property follows by considering the components cut off from v_t : they are either nested or disjoint. The union alternative cannot occur for the chosen smaller sides, since

$$\mu_n(S_e \cup S_{e'}) = \mu_n(S_e) + \mu_n(S_{e'}) - \mu_n(S_e \cap S_{e'}) < 1.$$

Here each of the first two terms is at most $1/2$, and the intersection contains v_t , whose mass is positive. Thus all components counted by $K_{M,t}$ are nested.

If the family is nonempty, let S be its largest component and choose a contour vertex immediately outside its interval. It has a representative at cyclic distance at most $2M + 2$ from t . The tree path from v_t to that vertex crosses all $K_{M,t}$ cut edges. Hence $K_{M,t}$ is at most the diameter of the horizontal contour image of the deterministic window of four times within $2M + 2$ of t . After the schedule has been fixed and holds deleted, this is a deterministic contour segment with at most $4M + 4$ horizontal steps. If the window covers the entire tour, the full horizontal contour instead has length $J \leq N \leq 4M + 4$. A segment with no horizontal steps has diameter zero; otherwise Lemma 6.4 applies. Uniformly in the schedule, we obtain

$$\mathbb{E}[K_{M,t}^2 \mid \text{schedule}] \leq CM. \quad (6.18)$$

Combining (6.17) and (6.18) proves the required bound on schedules outside $\mathcal{B}_{M,t}$ with balanced counts.

For the exceptional schedules, use a deterministic bound. Every interval counted by $K_{M,t}$ has its two delimiting tree traversals within cyclic distance $2M + 2$ of t . Distinct tree edges have distinct traversal occurrences, so $K_{M,t} \leq C(M + 1)$. Since $b_e \geq 1$, the same bound holds for $U_{M,t}$. By (6.16), the contribution of these schedules to its second moment is at most $C(M + 1)^4 e^{-cM}$, uniformly bounded in M . Finally, $U_{M,t} \leq N$ always. The unbalanced counts contribute at most $CN^2(N + 1)^3 e^{-cN}$ by (6.10), also uniformly bounded. This proves (6.15) under the full law. $\square$

6.5 Summing the scales

We now return to the inverse formula. Lemma 6.5 controls the cut sum at a fixed time; a grid converts this into a uniform bound over vertices.

Proof of Proposition 6.1. The one-vertex case has $G_n f = 0$, so set $A_n = 0$ there. Otherwise, for each dyadic $M \leq N/2$, choose a deterministic cyclic grid Γ_M . For $M \geq 8$, take its gaps at most $M/4$, its cardinality at most CN/M , and its distinct points separated by at least cM . For $M < 8$, use all step indices; the same cardinality and separation bounds hold with absolute constants.

An interval of length in $[M, 2M)$ containing a fixed time t contains a grid point within cyclic distance $2M$ of t . There are only a bounded number of grid points in this neighborhood. Consequently,

$$\sup_t U_{M,t} \leq C \max_{t \in \Gamma_M} U_{M,t}. \quad (6.19)$$

Every visit to a given vertex belongs to the same side of a tree cut. Also, counting the visits in each interval gives

$$\sum_{e \in \mathcal{E}_M} \frac{\mu_n(S_e)}{b_e} = \frac{1}{N} \sum_{t=0}^{N-1} U_{M,t}.$$

It follows that

$$W_M := \sup_{x \in V_n} \sum_{e \in \mathcal{E}_M} \frac{\mathbf{1}_{\{x \in S_e\}} + \mu_n(S_e)}{b_e} \leq C \max_{t \in \Gamma_M} U_{M,t}, \quad (6.20)$$

$$\mathbb{E}W_M \leq C \left(\sum_{t \in \Gamma_M} \mathbb{E}U_{M,t}^2 \right)^{1/2} \leq C \sqrt{N/M}. \quad (6.21)$$

No independence among the grid variables is needed here.

Fix f as in the proposition and write $m_f = \mu_n(|f|)$. For $e \in \mathcal{E}_M$, the source mass obeys

$$\left| \sum_{v \in S_e} \mu_n(v) f(v) \right| \leq \min(2M/N, m_f) \leq (2M/N)^{1-\rho} m_f^\rho. \quad (6.22)$$

Define, for the realized pair (M_n, T) ,

$$Y_n(M_n, T) = \sum_{\substack{M \leq N/2 \\ M \text{ dyadic}}} (2M/N)^{1-\rho} W_M.$$

By (6.21),

$$\mathbb{E}Y_n \leq C_\rho \sum_{\substack{M \leq N/2 \\ M \text{ dyadic}}} (M/N)^{1/2-\rho} \leq C_\rho. \quad (6.23)$$

The last bound is a geometric sum, using $\rho < 1/2$.

Conditional on M_n , the selected tree is uniform. Define

$$A_n(M_n) = \mathbb{E}[Y_n(M_n, T) \mid M_n]. \quad (6.24)$$

Apply Lemma 6.2 with $q = \mu_n f$. Taking absolute values, grouping by M , and using (6.22) proves

$$\|G_n f\|_\infty \leq A_n(M_n) m_f^\rho.$$

The variable in (6.24) is independent of the choice of f , so this bound holds simultaneously for all admissible functions. The tower property and (6.23) give $\sup_n \mathbb{E}A_n < \infty$, and Markov's inequality gives tightness. $\square$

The estimate depends only on the finite map law and its spanning trees. In particular, it controls the full potential of a small source, including any behavior on vertex sets that uniform approximation of continuous functions might fail to detect.

7 Resolvents and conditional path laws

Uniform convergence of energies concerns continuous functions, whereas the speed measure may be singular with respect to round area. The Green estimate bridges this difference: it makes a bounded source on a set of small quantum mass produce a uniformly small potential. After removing that potential, a resolvent is locally harmonic, so the compactness theorem for harmonic functions applies. We first give this argument for a deterministic sequence of graphs. The last subsection applies it in the coupled random environments and proves Theorem 1.1.

7.1 A deterministic convergence criterion

Let S^2 carry its round distance d_0 and area dA . Let μ be a nonatomic probability measure with full support. Suppose the quadratic form

$$\mathcal{E}(u, u) = \frac{1}{2} \int_{S^2} |\nabla u|^2 dA, \quad u \in C^\infty(S^2), \quad (7.1)$$

is closable in $L^2(\mu)$, and its closure $(\mathcal{E}, \mathcal{F})$ is a regular conservative Dirichlet form with an associated continuous-path diffusion B . We always start B with law μ . These hypotheses hold for the Liouville area measures used here; the time-change and Dirichlet-form constructions are treated in [8, 2], with general form theory in [7]. Our normalization is the explicit factor 1/2 in (7.1).

For each n , let $\mathcal{G}_n$ be a finite connected graph, allowing loops and multiple edges, with vertex set V_n and a strictly positive probability measure μ_n on V_n . Write

$$(H_n u)(v) = \sum_{vw} (u(v) - u(w)), \quad \mathbf{e}_n(u) = \sum_{\{v,w\}} (u(v) - u(w))^2.$$

The first sum counts incidences at v , and the second counts unoriented edges with multiplicity. Loop summands vanish. Let Y^n be the chain with generator

$$(\mathcal{L}_n u)(v) = -\frac{(H_n u)(v)}{2\mu_n(v)}. \quad (7.2)$$

It is reversible with respect to μ_n , and its form is $\mathcal{E}_n(u, u) = \mathbf{e}_n(u)/2$. Choose maps $\pi_n : V_n \rightarrow S^2$. Uniform comparison of functions always means

$$\|u_n - u \circ \pi_n\|_{\infty, V_n} \longrightarrow 0.$$

No injectivity of π_n is required.

Theorem 7.1 (From energies and Green bounds to path laws). *In the deterministic setting just defined, assume the following.*

- (i) $(\pi_n)_* \mu_n \Rightarrow \mu$, and the maximum projected edge length tends to zero:

$$\max_{\{v,w\} \in E(\mathcal{G}_n)} d_0(\pi_n(v), \pi_n(w)) \longrightarrow 0.$$

- (ii) Every uniformly convergent sequence with continuous limit u satisfies

$$\liminf_n \mathbf{e}_n(u_n) \geq \int |\nabla u|^2 dA,$$

where the right side is $+\infty$ unless $u \in W^{1,2}(S^2)$. Every smooth u has a uniformly convergent recovery sequence with energy tending to that integral.

- (iii) For every open $U \subset S^2$, sequences that are graph-harmonic at all vertices in $\pi_n^{-1}(U)$ and have uniformly bounded supremum and total energy are locally equicontinuous in U under π_n , and locally uniformly precompact there. Here equicontinuity means that, for every compact $K \subset U$,

$$\lim_{\delta \downarrow 0} \limsup_n \sup_{\substack{\pi_n(v), \pi_n(w) \in K \\ d_0(\pi_n(v), \pi_n(w)) \leq \delta}} |u_n(v) - u_n(w)| = 0,$$

and precompactness means uniform comparison with a continuous function on each compact subset after a common subsequence.

- (iv) For the centered solution operator

$$H_n G_n g = \mu_n g, \quad \mu_n(G_n g) = 0, \quad \mu_n(g) = 0,$$

there are constants $C < \infty$ and $\rho > 0$, independent of n , such that

$$\|G_n g\|_\infty \leq C \mu_n(|g|)^\rho \quad \text{if } \|g\|_\infty \leq 1 \text{ and } \mu_n(g) = 0. \quad (7.3)$$

Then the laws of $(\pi_n(Y_t^n))_{0 \leq t \leq T}$, with initial law μ_n , converge in $D([0, T], S^2)$ to the law of $(B_t)_{0 \leq t \leq T}$, for every finite T . The limit has continuous paths. In addition, for every $\alpha > 0$ and $f \in \tilde{C}(S^2)$, the resolvents converge uniformly under π_n .

We prove the theorem through four lemmas. Their hypotheses throughout are those of Theorem 7.1. Notice that full support and weak convergence of the measures already imply that $\pi_n(V_n)$ becomes dense: otherwise a fixed open ball of positive μ -mass would be missed along a subsequence.

Lemma 7.2 (Continuous Sobolev functions in the closed form). *If $u \in C(S^2) \cap W^{1,2}(S^2)$, then $u \in \mathcal{F}$ and its closed-form energy is the right side of (7.1). It has smooth approximations converging both uniformly and in the form norm. Two continuous versions of the same element of $L^2(\mu)$ are equal everywhere.*

Proof. In coordinate disks, mollification of a continuous Sobolev function converges both uniformly on smaller disks and in Sobolev norm. A smooth partition of unity gives smooth functions u_j on the sphere with these two convergences globally. Since μ is finite, uniform convergence also gives $L^2(\mu)$ convergence. Thus (u_j) is Cauchy for $\mathcal{E}(\cdot, \cdot) + \|\cdot\|_{L^2(\mu)}^2$ and has limit u in the stipulated closure. The gradient convergence identifies its energy. Polarization gives the same identity for the bilinear form on continuous Sobolev functions. Finally a nonzero continuous difference is bounded away from zero on some open set, which has positive μ -mass by full support. $\square$

Let $R_\alpha^n = (\alpha - \mathcal{L}_n)^{-1}$, and let R_α be the $L^2(\mu)$ resolvent of $(\mathcal{E}, \mathcal{F})$.

Lemma 7.3 (Uniform resolvent convergence). *For every $\alpha > 0$ and $f \in C(S^2)$, $R_\alpha f$ has a unique continuous version, which belongs to $W^{1,2}$, and*

$$\|R_\alpha^n(f \circ \pi_n) - (R_\alpha f) \circ \pi_n\|_\infty \longrightarrow 0. \quad (7.4)$$

The same conclusion holds with right-hand sides f_n converging uniformly under π_n to f .

Proof. Put $u_n = R_\alpha^n(f \circ \pi_n)$. The maximum principle and the resolvent equation give

$$\|u_n\|_\infty \leq \frac{\|f\|_\infty}{\alpha}, \quad H_n u_n = \mu_n g_n, \quad g_n = 2(f \circ \pi_n - \alpha u_n). \quad (7.5)$$

In particular g_n is uniformly bounded, and multiplying the equation by u_n shows that $e_n(u_n)$ is uniformly bounded.

We first prove equicontinuity. Fix a point $x \in S^2$, and choose disjoint open neighborhoods U of x and V of a different point. Take μ -null boundaries, with $\mu(V) > 0$. The neighborhood U may be made arbitrarily small in μ -mass because μ has no atoms. Write $U_n = \pi_n^{-1}(U)$ and $V'_n = \pi_n^{-1}(V)$, and set

$$q_n = g_n \mathbf{1}_{U_n} - \frac{\mu_n(g_n \mathbf{1}_{U_n})}{\mu_n(V'_n)} \mathbf{1}_{V'_n}.$$

For large n , the denominator is bounded below. Hence q_n has mean zero, bounded supremum, and $\mu_n(|q_n|) \leq C_1 \mu_n(U_n)$. Scaling (7.3) yields

$$\|G_n q_n\|_\infty \leq C_2 \mu_n(U_n)^\rho.$$

Also

$$e_n(G_n q_n) = \mu_n((G_n q_n) q_n) \leq \|G_n q_n\|_\infty \mu_n(|q_n|).$$

Thus $u_n - G_n q_n$ has bounded energy and supremum and is harmonic on U_n . Hypothesis (iii) makes this difference equicontinuous on compact subsets of U . Given an oscillation tolerance, first choose U so that $G_n q_n$ has smaller supremum, and then use harmonic equicontinuity inside U . A finite cover of S^2 proves equicontinuity of the original sequence. Boundedness and density of $\pi_n(V_n)$ now give uniformly convergent subsequences.

Let u be one such limit. Hypothesis (ii) and Lemma 7.2 show that $u \in \mathcal{F} \cap C(S^2)$. The discrete resolvent minimizes

$$J_n(v) = \frac{1}{2} e_n(v) + \alpha \mu_n(v^2) - 2\mu_n((f \circ \pi_n)v).$$

The energy lower bound, uniform convergence, and weak convergence of the measures imply

$$\mathcal{E}(u, u) + \alpha \mu(u^2) - 2\mu(fu) \leq \liminf_n J_n(u_n).$$

For any smooth v , take its uniform recovery sequence v_n . Minimality gives $J_n(u_n) \leq J_n(v_n)$, and the latter converges to $\mathcal{E}(v, v) + \alpha \mu(v^2) - 2\mu(fv)$. Smooth functions are a core of the closed form by its definition, so u minimizes this functional over all of $\mathcal{F}$. Strict convexity for $\alpha > 0$ identifies $u = R_\alpha f$ in $L^2(\mu)$. Full support makes its continuous version unique. Every subsequence has the same limit, which proves (7.4).

Finally, $\|R_\alpha^n a - R_\alpha^n b\|_\infty \leq \alpha^{-1} \|a - b\|_\infty$. This contraction bound proves the claim for varying right-hand sides. $\square$

7.2 Controlling small time intervals

Resolvent convergence integrates the walk over time. To recover the actual clock, we next construct functions that separate spatial sets and whose discrete generators are bounded. The needed continuum approximation is uniform, rather than merely in $L^2(\mu)$.

Lemma 7.4 (Uniform resolvent approximation). *For every $f \in C(S^2)$,*

$$\|\alpha R_\alpha f - f\|_\infty \longrightarrow 0 \quad \text{as } \alpha \longrightarrow \infty. \quad (7.6)$$

Proof. First suppose f is smooth, and put $w_\alpha = \alpha R_\alpha f$. The spectral theorem for the nonnegative self-adjoint form operator gives $w_\alpha \rightarrow f$ in the form norm. Indeed the spectral multiplier $\alpha/(\alpha + \lambda)$ tends to one and is bounded by one, so dominated convergence applies with spectral weight $1 + \lambda$. By Lemma 7.2,

$$\|\nabla(w_\alpha - f)\|_{L^2(dA)} \longrightarrow 0.$$

The maximum principle, or its discrete version followed by Lemma 7.3, gives

$$\min f \leq w_\alpha \leq \max f. \quad (7.7)$$

We also need convergence in round L^2 , which does not follow by changing the measure in the form norm. Let c_α be the round average of $w_\alpha - f$. Round Poincare gives

$$\|w_\alpha - f - c_\alpha\|_{L^2(dA)} \longrightarrow 0.$$

The constants c_α are bounded by (7.7). If a subsequence has limit c , then $w_\alpha \rightarrow f + c$ in round L^2 on that subsequence, and therefore along a further subsequence almost everywhere. Consequently $\min f \leq f + c \leq \max f$ almost everywhere for round area. Continuity of f makes its essential extrema equal to its extrema: every neighborhood of a minimizer or maximizer has positive round area. The two inequalities force $c \geq 0$ and $c \leq 0$. Thus $c = 0$, and $w_\alpha - f \rightarrow 0$ in round $W^{1,2}$.

Fix a small coordinate disk about a point x , and a fixed annulus inside that disk surrounding a smaller neighborhood of x . Radial averaging of the round $W^{1,2}$ error supplies, for each large α , a circle in this annulus on which the trace of $w_\alpha - f$ tends to zero in one-dimensional H^1 . The radii stay in a fixed compact subinterval of $(0, \infty)$, so the one-dimensional Sobolev inequality gives uniform convergence on these circles. Almost every radius has the Sobolev trace of the continuous representative, which is the trace we choose.

Let D_α be the disk inside the chosen circle. The resolvent equation is

$$\mathcal{E}(w_\alpha, v) + \alpha \int (w_\alpha - f)v \, d\mu = 0, \quad v \in \mathcal{F}. \quad (7.8)$$

Choose a constant K strictly above the boundary values of w_α and at least $\sup_{D_\alpha} f$. The positive part $(w_\alpha - K)^+$ on D_α , extended by zero, vanishes in a collar of the circle and is continuous Sobolev. Lemma 7.2 makes it an admissible test in (7.8). The two terms are nonnegative; the second is at least $\alpha \int ((w_\alpha - K)^+)^2 \, d\mu$. Thus $w_\alpha \leq K$ on the disk, first μ -almost everywhere and then everywhere by continuity and full support. The negative-part test gives the lower bound. Letting the strict slack tend to zero shows that the values inside the circle are bounded by its boundary values and the extrema of f in the disk.

The oscillation of f on sufficiently small disks is arbitrarily small. The boundary convergence and this comparison therefore give uniform convergence on a neighborhood of x . A finite cover proves (7.6) for smooth f . For continuous f , approximate it uniformly by a smooth function and use the contraction $\|\alpha R_\alpha\|_{\infty \rightarrow \infty} \leq 1$. $\square$

Lemma 7.5 (An exit estimate uniform over starting vertices). *For $d > 0$ and $v \in V_n$, let*

$$\tau_d^n(v) = \inf\{t \geq 0 : d_0(\pi_n(Y_t^n), \pi_n(v)) \geq d\}.$$

Then

$$\lim_{s \downarrow 0} \limsup_{n \rightarrow \infty} \sup_{v \in V_n} \mathbb{P}_v[\tau_d^n(v) \leq s] = 0. \quad (7.9)$$

The stationary projected path laws are tight in $D([0, T], S^2)$, and all their limits are supported on continuous paths.

Proof. Cover the sphere by finitely many round disks of radius less than $d/8$. For each choose a smooth function f in $[0, 1]$, zero on that disk, and one outside the concentric disk of radius $d/2$. If $\pi_n(v)$ is in the smaller disk and displacement from $\pi_n(v)$ reaches d , the value of f at the new location is one.

Fix $b > 0$. Lemmas 7.4 and 7.3, first with a sufficiently large fixed α and then with sufficiently large n , give for each of this finite family a function

$$v_n = \alpha R_\alpha^n(f \circ \pi_n), \quad \|v_n - f \circ \pi_n\|_\infty \leq b.$$

Its generator satisfies

$$\mathcal{L}_n v_n = \alpha(v_n - f \circ \pi_n), \quad \|\mathcal{L}_n v_n\|_\infty \leq C_b,$$

with a common C_b for the finite family and all sufficiently large n . Start the chain at v in the corresponding small disk and stop at $\sigma = \tau_d^n(v) \wedge s$. If $p = \mathbb{P}_v[\tau_d^n(v) \leq s]$, then

$$\mathbb{E}_v[v_n(Y_\sigma^n)] \geq p - b, \quad \mathbb{E}_v[v_n(Y_\sigma^n)] = v_n(v) + \mathbb{E}_v \int_0^\sigma \mathcal{L}_n v_n(Y_t^n) dt \leq b + C_b s.$$

Hence $p \leq 2b + C_b s$, uniformly over all starting vertices. First let $s \downarrow 0$ and then $b \downarrow 0$ to obtain (7.9).

Apply the strong Markov property to the chain Y^n after any stopping time. Since (7.9) is uniform over its current vertex, it gives the usual stopping-time increment condition for the projected paths. The projection itself need not be Markov. Apply Aldous's stopping-time criterion [1] to the three coordinate functions of the round sphere in $\mathbb{R}^3$; their values are bounded. This gives tightness of each coordinate path law. The largest coordinate jump is bounded by the maximum projected edge length, which tends to zero. Under J_1 convergence any jump of a limit is approximated by jumps of the approximating paths, so every coordinate limit is continuous. Coordinate tightness with continuous limits gives joint tightness with continuous limits in $\mathbb{R}^3$: on continuous limits, J_1 convergence agrees with uniform convergence, allowing a common time comparison for the three coordinates. The limits remain in the closed set S^2 , proving the assertion. $\square$

7.3 Identifying the entire conditional law

Completion of the proof of Theorem 7.1. Take a subsequence of the stationary projected path laws with a weak limit. By Lemma 7.5, diagonal extraction on the horizons $1, 2, \dots$ gives a consistent continuous-path law on $[0, \infty)$, denoted by Q . Restrictions are consistent because the limits are continuous at the restriction times.

Let $f_0, \dots, f_k \in C(S^2)$ and $\alpha_1, \dots, \alpha_k > 0$. Integrate the stationary finite-dimensional expectation at times $0, t_1, t_1 + t_2, \dots, t_1 + \dots + t_k$ against $\prod_{j=1}^k e^{-\alpha_j t_j} dt_j$. The Markov property and the resolvent identity give exactly

$$\mu_n \left[(f_0 \circ \pi_n) R_{\alpha_1}^n \left((f_1 \circ \pi_n) R_{\alpha_2}^n \left(\dots (f_{k-1} \circ \pi_n) R_{\alpha_k}^n (f_k \circ \pi_n) \dots \right) \right) \right]. \quad (7.10)$$

Successive applications of Lemma 7.3, including its varying-data assertion, and weak convergence of the measures show that (7.10) converges to

$$\mu[f_0 R_{\alpha_1} (f_1 R_{\alpha_2} (\cdots f_{k-1} R_{\alpha_k} f_k \cdots))].$$

By the semigroup-resolvent relation for the closed form, this is the same integrated expectation for the stationary diffusion B .

It is also the integrated expectation for Q . On bounded time boxes this follows from weak path convergence, since evaluation at fixed times is continuous at continuous paths. Dominated convergence then integrates over the gaps. The contribution of the complement of a large time box is bounded uniformly by the tails of the product of exponentials, times $\prod_j \|f_j\|_\infty$.

The product expectations under Q and under B are bounded continuous functions of $(t_1, \dots, t_k)$, including zero gaps. Uniqueness of multivariate Laplace transforms makes these functions equal: one may apply one-dimensional uniqueness successively in each coordinate, and then use continuity to remove almost-everywhere qualifications. Product tests determine probability measures on the compact product $(S^2)^{k+1}$. Thus Q and the law of B have the same finite-dimensional distributions, and hence the same law on continuous paths. Every subsequential limit is the same, proving the theorem. $\square$

7.4 The clock and the metric-space conclusion

We now return to the random maps. The exact relation between the graph energy and the prescribed walk fixes the clock without any subsequence-dependent factor.

Lemma 7.6 (Exact energy of the accelerated walk). *For the half-edge walk of Theorem 1.1,*

$$-\langle u, nL_n u \rangle_{L^2(\mu_n)} = \frac{1}{2} \mathbf{e}_n(u).$$

Equivalently, $nL_n = -\frac{1}{2}\mu_n^{-1}H_n$, where μ_n^{-1} denotes division by vertex mass.

Proof. The generator convention gives $H_n = -\deg L_n$. Since $\mu_n(v) = \deg(v)/(2n)$,

$$-\sum_v \mu_n(v) u(v) nL_n u(v) = \frac{1}{2} \sum_v u(v) H_n u(v) = \frac{1}{2} \sum_{\{v,w\}} (u(v) - u(w))^2.$$

Loops contribute zero to both sides. Their degree contributions are already included in the first identity, so no holding-time correction is needed. $\square$

Proof of Theorem 1.1. Start with any subsequence. Retain the random variable A_n of Proposition 6.1 together with all the contour, spatial, and local and whole-sphere array variables used in Theorem 5.4 and Proposition 5.5. Its uniform first-moment bound gives tightness, so these variables admit a joint further extraction. Apply the sphere results to this enlarged extraction and use a Skorokhod representation. On the resulting probability-one event, their conclusions hold and A_n converges to a finite random variable. In particular $\sup_n A_n < \infty$ after discarding finitely many initial terms.

Fix a realization in this probability-one event. Its limiting quantum sphere is now a fixed measured conformal sphere. The projections π_n , measures, edge mesh, uniform energy bounds and recovery, and harmonic compactness satisfy hypotheses (i)–(iii) of Theorem 7.1. The realized bounded A_n gives hypothesis (iv). The Liouville form is exactly (7.1) with $\mu = \mu_h$. By Lemma 7.6,

the chain in that deterministic theorem is $Y_t^n = X_{nt}^n$. Consequently its conditional projected path law converges to the conditional law of B^h , for every fixed finite horizon. All probabilities in this invocation are taken on the individual fixed graphs. The random object being compared is the conditional probability law itself.

It remains to express this comparison in the metric δ_T . The round topology and the D_h topology agree on the compact limiting sphere, so the identity map between them is uniformly continuous in both directions. The path-law convergence just proved therefore also holds in $D([0, T], (S^2, D_h))$.

The correspondences in Proposition 5.5 have distortions tending to zero. The standard correspondence gluing constructs a common metric space containing isometric copies of all $(V_n, a_n d_n)$ and (S^2, D_h) , with

$$\sup_{v \in V_n} \text{dist}_Z(v, \pi_n(v)) \longrightarrow 0.$$

For completeness, for each n attach its copy to the sphere by cross-distances obtained by taking the infimum over corresponding pairs, adding numbers $\varepsilon_n \downarrow 0$ larger than half the respective distortions; use the shortest-path metric on the union, identifying only the common sphere, and then take the completion. This preserves the metrics of every copy. The images become close in Hausdorff distance, and their measures become close in Prokhorov distance.

Apply the embedding to each coordinate of a chain path. Its distance in the common Skorokhod space from its π_n -projected path is bounded by the displayed supremum, using the identity time change. The corresponding probability laws therefore have Prokhorov distance tending to zero. Combining this with the projected law convergence shows that all three terms defining δ_T tend to zero in the coupled extraction.

We have obtained almost-sure convergence in a further coupling from every original subsequence, to the prescribed quantum sphere and its conditional diffusion law. The subsequence principle gives convergence in distribution through all positive integers n . Forgetting the root, the spanning tree, and the contour marks leaves precisely the states in Theorem 1.1. The same choice $c = 1$ works for every time horizon. $\square$

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